Highlights

Multi-Task Particle Swarm Optimization with Knowledge-Guided Elite Refinement and Two-Phase Weight Scheduling for High-Dimensional Feature Selection

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- Closed-loop knowledge-guided swarm learning for high-D feature selection.
- MI seeding with adaptive recovery prevents stagnation and restores diversity.
- Knowledge-guided elite refinement accumulates feature signals with decay.
- Two-phase weight scheduling decouples feature pruning from accuracy tuning.
- KDS-EL achieves 95.74% accuracy with only 87.8 features on 20 benchmarks.

Multi-Task Particle Swarm Optimization with Knowledge-Guided Elite Refinement and Two-Phase Weight Scheduling for High-Dimensional Feature Selection

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ABSTRACT

Feature selection in high-dimensional data remains a central challenge for evolutionary computation. Particle swarm optimization suffers from three structural deficiencies: random initialization provides insufficient guidance in sparse high-dimensional spaces, uniform feature perturbation fails to exploit cross-iteration feature-importance signals, and static objective preferences cannot adapt to different evolutionary phases. To address these limitations, we propose KDS-EL (Knowledge-Guided Dual-Phase Swarm Evolutionary Learning), a knowledge-guided evolutionary framework with a closed-loop feature-importance learning mechanism. First, to overcome inefficient initialization, we propose mutual-information-guided initialization with adaptive re-initialization, injecting prior knowledge and recovering population diversity when stagnation occurs. Second, to exploit discarded cross-iteration feature-importance signals, we propose a knowledge-guided elite refinement strategy that accumulates feature-importance knowledge and steers elite solutions toward historically effective features. Third, to align objective weights with search phases, we propose two-phase weight scheduling that emphasizes early dimensionality reduction and late-stage accuracy refinement. On 20 high-dimensional benchmarks, KDS-EL achieves the best classification performance among all competitors, with a mean accuracy of 0.9574 and only 87.8 selected features, outperforming the strongest single- and multi-objective competitors by +3.97 and +4.42 percentage points; Wilcoxon signed-rank tests with Holm–Bonferroni correction confirm statistically significant improvements over all 13 competitors.

1. Introduction

High-dimensional data have emerged as a fundamental challenge in classification problems across diverse domains, including genomics, text mining, and computer vision. These datasets typically contain hundreds or even tens of thousands of features, while only a limited number of labeled samples are available, resulting in the well-known $p \gg n$ scenario. Under such conditions, conventional learning algorithms often suffer from severe overfitting, increased computational complexity, and reduced interpretability due to the curse of dimensionality [1, 2]. Feature selection (FS) has therefore become an essential preprocessing technique that aims to identify a compact and informative subset of features while preserving classification performance. However, selecting an optimal feature subset remains highly challenging because the search space grows exponentially with the number of features D, and the underlying subset-selection problem is NP-hard [3]. Moreover, individual feature relevance measures are insufficient to characterize complex feature interactions and dependencies, which are often critical for achieving robust classification performance.

FS methods are generally categorized into filter, wrapper, and embedded approaches [1, 4]. Filter methods rank features according to statistical criteria and offer high computational efficiency, but they usually evaluate features independently of the classifier and may overlook feature interactions relevant to classification. Embedded methods incorporate feature selection into the learning process, providing a compromise between selection efficiency and predictive performance; however, their effectiveness is often dependent on the characteristics of specific learning models. Wrapper methods directly optimize feature subsets according to classifier performance and can achieve competitive classification results, but the repeated model evaluations make them computationally expensive, especially in high-dimensional search spaces. To address the combinatorial complexity of wrapper-based FS, evolutionary computation has been extensively investigated as a promising optimization paradigm [3, 5–7]. By maintaining a population of candidate solutions, evolutionary algorithms can perform global exploration of feature subsets without requiring gradient information. Among them, particle swarm optimization (PSO), genetic algorithms, and differential evolution have demonstrated promising results on high-dimensional FS tasks.

PSO has become one of the most widely adopted evolutionary algorithms for FS due to its simple structure, efficient information-sharing mechanism, and fast convergence. Nevertheless, conventional PSO-based FS methods still encounter three structural limitations when dealing with sparse and high-dimensional search spaces. First, random initialization distributes particles without prior guidance throughout the binary search space, whereas effective feature subsets usually occupy only a small region with limited active dimensions. Consequently, a considerable number of fitness evaluations may be wasted on redundant and suboptimal solutions before the swarm identifies promising search regions. Second, conventional uniform feature perturbation strategies treat all dimensions equally and neglect the feature-level improvement information accumulated during previous evaluations, preventing the search from exploiting historically informative features. Third, existing PSO-based FS methods typically employ fixed objective weights to balance classification error and feature subset size during the entire optimization process. However, the desired trade-off between feature reduction and accuracy refinement varies across search stages: aggressive sparsity enhancement is preferred in early exploration, whereas accurate decision boundary refinement becomes more important after a compact subset has been obtained.

Recent evolutionary FS studies have attempted to improve search performance through enhanced population initialization, adaptive evolutionary operators, and refined fitness evaluation strategies [8–11]. Although these approaches have achieved promising results, they usually focus on isolated components of the optimization process and do not jointly address inefficient initialization, insufficient exploitation of accumulated feature knowledge, and inflexible objective weighting. Recently, multi-task PSO has emerged as an effective paradigm for high-dimensional FS with fixed-length binary representations. For instance, MEL [12] adopts cooperative subpopulations and a fixed-length representation to efficiently handle feature spaces containing tens of thousands of dimensions. Despite its effectiveness, existing multi-task PSO-based FS frameworks still leave several challenges unresolved: the initial population is generally generated without sufficient feature-level prior knowledge, useful feature-importance information discovered during evolution is not explicitly retained and exploited, and the balance between classification performance and feature sparsity remains unchanged across different search stages.

To address these limitations, this paper proposes Knowledge-Guided Dual-Phase Swarm Evolutionary Learning (KDS-EL), a closed-loop framework that cohesively integrates domain prior injection, cross-iteration knowledge accumulation, and dynamic objective scheduling into a cooperative multi-task PSO backbone.

The main contributions of this paper are summarized as follows:

1. *MI-Guided Initialization with Adaptive Re-initialization*: A hybrid seeding strategy that injects continuous feature-relevance priors into population construction to eliminate early-stage search waste, coupled with an adaptive re-initialization mechanism that restores swarm diversity upon evolutionary stagnation (Section 3.2).
2. *Knowledge-Guided Elite Refinement (KER)*: A targeted elite perturbation operator that dynamically harvests cross-iteration feature-importance feedback with exponential memory decay, directing elite solutions toward historically effective features within a strictly bounded evaluation overhead (Section 3.3).

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Table 1: Methodological taxonomy of investigated evolutionary feature selection algorithms.

		Knowledge Engineering			Search Control		
Algorithm	Paradigm	MI Prior Seeding	Online Memory	Memory Decay	Elite Refinement	Dynamic Weighting	Complexity
BPSO [7]	SO-SI	×	×	×	×	×	High
FWPSO [8]	SO-SI	○	×	×	×	×	Moderate
SaWDE [9]	SO-DE	×	×	×	×	○	High
IGPSO [10]	SO-SI	○	×	×	×	×	Moderate
PSO-CSM [11]	SO-SI	○	×	×	×	×	Moderate
NSGA-II-FS [13]	MO-GA	×	×	×	×	×	High
MOEA/D-FS [14]	MO-EA	×	×	×	×	×	High
MOPSO-FS [15]	MO-SI	×	×	×	×	×	High
fMOPSO-FS [16]	MO-SI	×	×	×	×	×	High
W-MOPSO-FS [17]	MO-SI	○	×	×	×	×	High
MEL [12]	MT-SI	×	✓	×	×	×	High
KDS-EL (Proposed)	**MT-SI**	✓	✓	✓	✓	✓	**Low**

✓: Fully supported; ○: Partially supported; ×: Absent / not supported.
Paradigms: SO: Single-Objective; MO: Multi-Objective; MT: Multi-Task; SI: Swarm Intelligence; DE: Differential Evolution; GA: Genetic Algorithm; EA: Evolutionary Algorithm.

3. *Two-Phase Weight Scheduling (TPS)*: A stage-aware objective scheduling mechanism that temporally decouples early aggressive feature pruning from late-stage accuracy refinement, overcoming the rigidity of static weighting (Section 3.4).
4. *Comprehensive Empirical Validation*: Extensive evaluations across 20 high-dimensional benchmark datasets (spanning 500 to 54,675 features) against 13 state-of-the-art competitors, rigorously validated through non-parametric statistical testing, component contribution analysis, and parametric sensitivity investigations (Section 4).

Extensive experiments demonstrate that KDS-EL achieves the best classification performance, obtaining an average classification accuracy of 0.9574 with only 87.8 selected features. Compared with the strongest single-objective and multi-objective competitors, KDS-EL improves accuracy by 3.97 and 4.42 percentage points, respectively, while achieving a 3.4× speedup over standard MEL [12] (reducing average runtime from 175.0 s to 51.4 s). Furthermore, Wilcoxon signed-rank tests with Holm–Bonferroni correction confirm that the improvements are statistically significant over all 13 competitors.

The remainder of this paper is organized as follows. Section 2 formulates the feature selection optimization problem and reviews related literature and background concepts. Section 3 presents the proposed KDS-EL framework and formulates its three constituent mechanisms in detail. Section 4 presents comprehensive experimental results, statistical tests, component contribution analysis, and parameter sensitivity investigations. Finally, Section 5 concludes the paper and discusses prospective research directions.

2. Related Work and Background

This section establishes the formal mathematical foundation of feature selection and systematically reviews the relevant evolutionary computation paradigms. Section 2.1 first formalizes the combinatorial optimization problem and categorizes classical search paradigms. Section 2.2 subsequently analyzes evolutionary swarm mechanics alongside the structural trade-offs between weighted scalarization and multi-objective Pareto formulations. To address high-dimensional search challenges, Section 2.3 scrutinizes the cooperative multi-task evolutionary learning framework and pinpoints its operational bottlenecks. Finally, Section 2.4 investigates knowledge-guided evolutionary search, contrasts offline filter priors against online closed-loop memory, and establishes the taxonomic positioning of KDS-EL in contrast to existing literature.

2.1. Problem formulation and search paradigms in feature selection

The essence of feature selection lies in combinatorial optimization, aiming to identify an optimal, minimal subset of informative features from a high-dimensional feature space to maximize classification performance while eliminating redundant and uninformative dimensions [1, 2]. Formally, let $\mathcal{D} = \{(\mathbf{f}_i, y_i)\}_{i=1}^{N_{\text{samp}}}$ denote a supervised training dataset comprising N_{samp} instances, where $\mathbf{f}_i \in \mathbb{R}^D$ represents the full feature vector in the original D-dimensional space and $y_i \in \{1, 2, \ldots, C\}$ designates the discrete class label across C categories. In discrete binary representations, a candidate feature subset is formulated as a binary decision indicator vector $\mathbf{x} \in \{0, 1\}^D$, defined as:

$$x_j = \begin{cases} 1, & \text{if feature } f_j \text{ is selected,} \\ 0, & \text{if feature } f_j \text{ is discarded.} \end{cases} \tag{1}$$

The primary objective of feature selection is to simultaneously minimize the classification error rate and the cardinality of the selected feature subset. In evolutionary wrapper feature selection, this multi-criterion goal is mathematically formalized as minimizing a scalarized single-objective fitness function $f(\mathbf{x})$ [18, 19]:

$$\min_{\mathbf{x}\in\{0,1\}^D} f(\mathbf{x}) = w_a \cdot \text{ErrorRate}(\mathbf{x}) + w_f \cdot \left(\frac{\|\mathbf{x}\|_0}{D}\right), \tag{2}$$

subject to $\mathbf{x} \in \{0, 1\}^D$ and $w_a + w_f = 1$, where ErrorRate($\mathbf{x}$) represents the cross-validation classification error evaluated on the masked feature subset $\{\mathbf{f}_i \odot \mathbf{x}\}$ using an inductive k-nearest neighbors (k-NN) learner, with $\odot$ denoting the Hadamard element-wise product. The term $\|\mathbf{x}\|_0$ denotes the number of active features, $\|\mathbf{x}\|_0/D$ represents the feature selection ratio, and $w_a, w_f \in [0, 1]$ serve as trade-off coefficients balancing predictive accuracy against subset compactness. Because the cardinality of the discrete search space is 2^D, evaluating all candidate subsets exhaustively is NP-hard and computationally intractable when $D \gg 10^3$. Traditional feature selection techniques are broadly categorized into three classical paradigms based on their evaluation criteria: filter, embedded, and wrapper methods [2, 4]. Specifically, *filter methods* evaluate feature subsets based solely on intrinsic statistical or information-theoretic metrics—such as mutual information in mRMR [20], correlation in FCBF [21], or centrality scores [22]—independently of any classification algorithm. While filter methods are computationally lightweight with $O(DN_{\text{samp}} + D \log D)$ complexity, their classifier-agnostic evaluation and rigid hard truncation permanently discard weakly relevant features that exhibit strong collective synergy when combined. Conversely, *embedded methods* perform feature selection natively during classifier training by embedding sparsity constraints into model parameter estimation, such as L_1-regularized regression in Bolasso [23] or tree-based ensemble importance in FS-GBDT [24]. Although embedded methods bridge computational efficiency and predictive feedback, they remain architecturally bound to specific learners and frequently encounter performance degradation under severe sample scarcity where $N_{\text{samp}} \ll 100$. In contrast, *wrapper methods* utilize a target learning algorithm as a black-box evaluator, assessing candidate feature masks directly through cross-validation predictive error feedback [4, 18]. By tightly coupling subset evaluation with classifier feedback, wrappers naturally capture complex non-linear feature interactions and yield superior classification accuracy.

To navigate the discrete 2^D combinatorial space in wrapper feature selection without gradient information, Evolutionary Computation paradigms—including Genetic Algorithms [25], Ant Colony Optimization [26], Artificial Bee Colony [27], and Swarm Intelligence [3, 28]—have emerged as dominant search heuristics. To combat the curse of dimensionality in small-sample high-dimensional data, various advanced single-objective evolutionary strategies have been introduced. For instance, self-adaptive differential evolution with adaptive weighting (SaWDE) [9] dynamically

balances classification error against subset cardinality; feature-weight particle swarm optimization (FWPSO) [8] injects offline feature weights to steer velocity updates; importance-guided particle swarm optimization (IGPSO) [10] employs multi-layer perceptron importance scores to bias particle movements; and particle swarm optimization with comprehensive scoring mechanism (PSO-CSM) [11] leverages multi-criterion feature evaluation to restrict search trajectories. Despite these advancements, standard single-objective evolutionary wrappers face severe structural bottlenecks in ultra-high-dimensional spaces where $D \gg 10^3$. Uniform random initialization scatters individuals indiscriminately across the search space, generating overly dense candidate masks with approximately $D/2$ active features that burden classifier evaluations and provoke premature overfitting. Furthermore, stochastic variation operators treat feature perturbations as memoryless trials, discarding historical feature-importance feedback across generations. Finally, static or coarsely tuned objective weighting enforces rigid trade-offs throughout evolution, failing to adapt to the distinct requirements of early-stage dimensional reduction and late-stage accuracy refinement.

2.2. Evolutionary swarm intelligence and multi-objective formulations

Standard particle swarm optimization, originally introduced by Kennedy and Eberhart [5], is a population-based stochastic meta-heuristic inspired by the collective foraging behaviors of bird flocks and fish schools. A swarm consists of N candidate particles navigating a D-dimensional continuous solution space $\mathbb{R}^D$. At iteration t, each particle i is characterized by a continuous position vector $\mathbf{p}_i^{(t)} \in \mathbb{R}^D$ and an associated velocity vector $\mathbf{v}_i^{(t)} \in \mathbb{R}^D$, steered dynamically by its personal historical best position $\mathbf{p}_{\text{best},i} \in \mathbb{R}^D$ and the swarm's global best position $\mathbf{g}_{\text{best}} \in \mathbb{R}^D$. The velocity and position updates of particle i at iteration $t+1$ are formulated as follows [6]:

$$\mathbf{v}_i^{(t+1)} = \omega \mathbf{v}_i^{(t)} + c_1 r_1 (\mathbf{p}_{\text{best},i} - \mathbf{p}_i^{(t)}) + c_2 r_2 (\mathbf{g}_{\text{best}} - \mathbf{p}_i^{(t)}), \tag{3}$$

$$\mathbf{p}_i^{(t+1)} = \mathbf{p}_i^{(t)} + \mathbf{v}_i^{(t+1)}, \tag{4}$$

where $\omega \in [0, 1]$ denotes the inertia weight controlling momentum retention, $c_1, c_2 > 0$ are the cognitive and social acceleration coefficients governing the attraction toward individual and collective best experiences, and $r_1, r_2 \sim \mathcal{U}(0, 1)$ are independent uniform random factors providing stochastic exploration capability.

To adapt continuous PSO to the combinatorial nature of binary feature selection where candidate solutions are represented as binary indicator masks $\mathbf{x}_i \in \{0, 1\}^D$, Kennedy and Eberhart [7] proposed Binary PSO (BPSO). In BPSO, continuous velocities are mapped into selection probabilities via a Sigmoid transfer function:

$$S(v_{i,j}^{(t+1)}) = \frac{1}{1 + \exp(-v_{i,j}^{(t+1)})}, \tag{5}$$

and the corresponding binary decision variable $x_{i,j}^{(t+1)}$ is sampled probabilistically as:

$$x_{i,j}^{(t+1)} = \begin{cases} 1, & \text{if } r_{i,j} < S(v_{i,j}^{(t+1)}), \\ 0, & \text{otherwise}, \end{cases} \tag{6}$$

where $r_{i,j} \sim \mathcal{U}(0, 1)$ is an independent uniform random threshold. Although BPSO exhibits fast convergence on low-dimensional benchmarks, it encounters severe exploration-exploitation dilemmas when applied to ultra-high-dimensional biomedical domains where $D \gg 10^3$. Because the Sigmoid function satisfies $S(0) = 0.5$, continuous velocities hovering near zero inevitably yield an expected active feature density of approximately $D/2$. In typical biomedical datasets containing $D \geq 10{,}000$ dimensions, this stochastic mapping generates candidate subsets with more than 5,000 active features, heavily overloading wrapper classifier evaluations with $O(N_{\text{samp}}^2 d_{\text{selected}})$ complexity and triggering severe overfitting under small sample sizes where $N_{\text{samp}} \ll 100$. Furthermore, continuous stochastic sampling introduces high generational variance: even when velocity vectors stabilize, candidate feature masks continue to jitter stochastically, causing particle trajectories to degrade into memoryless random walks across the combinatorial space.

Because feature selection intrinsically entails two conflicting objectives—minimizing classification error and minimizing the number of selected features—Multi-Objective Feature Selection (MOFS) algorithms model this trade-off through Pareto optimality. Pioneered by Xue et al. [17], diverse multi-objective evolutionary algorithms have been tailored for feature selection, encompassing genetic algorithms like NSGA-II-FS [13], decomposition-based heuristics like MOEA/D-FS [14], canonical swarm methods including MOPSO-FS [15], filter–wrapper hybrids such as W-MOPSO-FS [17], and fuzzy Pareto-dominance techniques such as fMOPSO-FS [16]. Despite their theoretical capability to generate a non-dominated Pareto front, MOFS methods encounter two critical operational bottlenecks under ultra-high-dimensional and sample-scarce regimes where $D \gg 10^3$ and $N_{\text{samp}} \ll 100$. First, maintaining an external non-dominated archive and executing Pareto ranking over dense high-dimensional vectors incurs substantial computational overhead, often requiring 110–140 seconds of wall-clock execution time (Section 4.2.6). Second, under severe sample sparsity, Pareto dominance and fuzzy ranking criteria fail to exert adequate selection pressure toward extreme feature sparsity; in ultra-high dimensions, candidate solutions become mutually non-dominated across irrelevant feature dimensions, populating the Pareto front with non-sparse subsets that retain thousands of active features rather than the compact biomarker signatures with fewer than 100 features required for clinical interpretability. To circumvent Pareto archive bloat and achieve extreme subset sparsity within tight computational budgets, weighted single-objective scalarization remains the prevailing formulation in high-dimensional biomarker discovery, as exemplified by SaWDE [9], FWPSO [8], and PSO-CSM [11]. However, existing weighted frameworks enforce static, inflexible trade-off weights throughout the optimization process, creating a fundamental evolutionary phase misalignment: the search process requires aggressive feature reduction during early exploration to eliminate vast noise subspaces, whereas delicate decision-boundary fine-tuning is required in late-stage exploitation to maximize classification accuracy.

2.3. Multi-task evolutionary learning and the MEL framework

Evolutionary Multi-Task Optimization (EMTO) aims to solve multiple distinct optimization tasks simultaneously by facilitating inter-task knowledge transfer through a unified chromosomal representation [29]. In feature selection, EMTO frameworks partition the global population into collaborative subpopulations to explore complementary search subspaces, including minion-assisted multi-tasking [30], evolutionary multi-task PSO for hyperspectral and gene expression data [31, 32], multi-task bee colony band selection [33], and large-scale multi-objective multi-tasking [34]. Most notably, Zhang et al. [12] introduced Multi-Task Evolutionary Learning (MEL), which establishes a dual-subpopulation PSO framework. In MEL, a population of N particles is partitioned into two collaborative subpopulations $\mathcal{P}_1$ and $\mathcal{P}_2$, with each subpopulation comprising $N/2$ individuals operating on complementary search tasks. Specifically, *Subpopulation 1* executes continuous swarm exploration, updating continuous velocities and positions using PSO dynamics augmented with an inter-task knowledge transfer channel directing particles toward the historical global best of Subpopulation 2 ($\mathbf{g}_{\text{best},2} \in \mathbb{R}^D$):

$$\mathbf{v}_{1,i}^{(t+1)} = \omega \mathbf{v}_{1,i}^{(t)} + c_1 r_1 (\mathbf{p}_{\text{best},1,i} - \mathbf{p}_{1,i}^{(t)}) + c_2 r_2 (\mathbf{g}_{\text{best},1} - \mathbf{p}_{1,i}^{(t)}) + c_3 r_3 (\mathbf{g}_{\text{best},2} - \mathbf{p}_{1,i}^{(t)}), \tag{7}$$

where $\omega \in [0, 1]$ is the inertia weight, $c_1, c_2, c_3 > 0$ are acceleration coefficients governing self-cognition, social learning, and cross-task transfer, and $r_1, r_2, r_3 \sim \mathcal{U}(0, 1)$ are independent stochastic factors. Concurrently, *Subpopulation 2* performs importance-guided discrete sampling directly from a shared feature-importance matrix $\mathbf{W}_{\text{MEL}} \in \mathbb{R}^D$, whose elements accumulate the historical selection frequencies of features across successful candidate solutions.

Both subpopulations evaluate candidate subsets using a shared scalar fitness function similar to Eq. (2) that combines classification error and feature subset ratio via fixed static trade-off weights $\alpha, \beta \in [0, 1]$ satisfying $\alpha + \beta = 1$.

While MEL demonstrates the utility of cross-task knowledge transfer, its search dynamics in ultra-high-dimensional spaces where $D \gg 10^3$ suffer from critical structural bottlenecks. Continuous particles in $\mathcal{P}_1$ are initialized uniformly across the unit hypercube, lacking domain-informed guidance toward sparse biomarker subspaces and wasting substantial evaluation budget exploring irrelevant dimensions in early iterations. Moreover, the importance matrix $\mathbf{W}_{\text{MEL}}$ accumulates historical feature frequencies unconditionally without exponential forgetting or elite fitness verification, allowing obsolete early choices to permanently bias subsequent discrete sampling in $\mathcal{P}_2$ and trapping the swarm in local optima. In addition, rigid static weighting prevents the algorithm from dynamically adjusting its selection pressure between early aggressive feature pruning and late high-precision boundary refinement. The proposed KDS-EL framework directly adopts this dual-subpopulation multi-task foundation while resolving each of these structural deficiencies through targeted mechanisms.

2.4. Knowledge-guided search and taxonomic positioning

Integrating domain-specific prior knowledge into meta-heuristics has attracted surging attention as an effective means to escape the blindness of black-box search [19, 35, 36]. In high-dimensional feature selection, Mutual Information (MI), rooted in Shannon's information theory [37], serves as the primary information-theoretic metric to quantify non-linear feature relevance. For a fea-

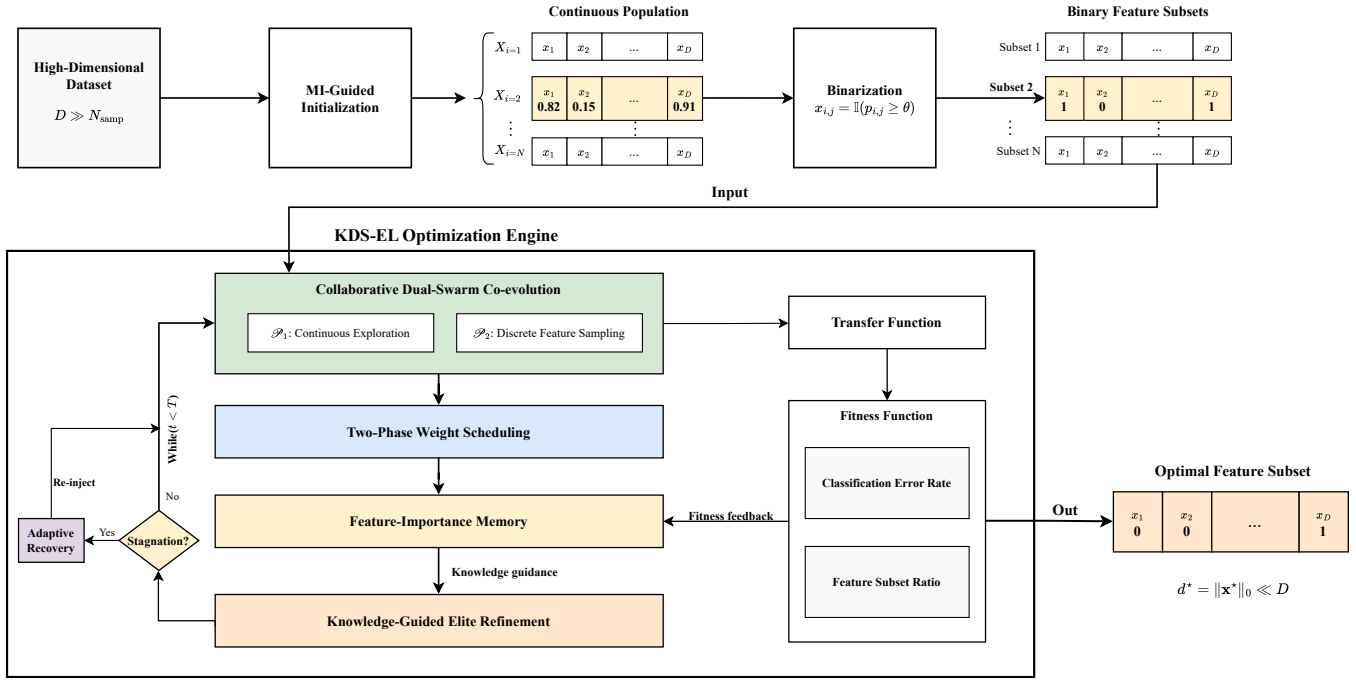


Figure 1: Example of the feature selection process via KDS-EL.

ture variable f_j and the discrete class label $Y \in \{1, \ldots, C\}$, the mutual information $I(f_j; Y)$ is defined as:

$$I(f_j; Y) = \sum_{v \in \text{Val}(f_j)} \sum_{c=1}^{C} P(f_j = v, Y = c) \log_2 \frac{P(f_j = v, Y = c)}{P(f_j = v)P(Y = c)}, \tag{8}$$

which directly measures the shared information between feature f_j and class labels Y, with high values indicating strong discriminative capability. Conventional filter feature selection methods [19, 20] evaluate mutual information across all D dimensions and apply a rigid hard truncation that retains only the top-k ranked features, permanently discarding the remainder. Despite its computational speed of $O(DN_{\text{samp}} + D \log D)$, hard truncation fundamentally ignores multi-feature synergy: individual features with moderate scores that are collectively decisive when combined are permanently eliminated from the search space.

Existing knowledge-guided evolutionary algorithms typically adopt one-shot offline injection via either static filter pre-filtering or static velocity biasing. In static filter pre-filtering, exemplified by MI-MPODE [36] and NNS-DFPG-CSA [19], algorithms compute feature rankings offline and initialize the population or restrict the search space to the top-k ranked features. In static velocity biasing, such as FWPSO [8], static feature-weight vectors are incorporated to bias particle velocity equations throughout evolution. While beneficial, one-shot knowledge injection lacks online adaptability: once injected, the static prior is never updated or refined by the empirical trial-and-error feedback generated during subsequent wrapper evaluations. Conversely, learning-based surrogates, such as neural networks [10] or Gaussian processes [38], introduce severe surrogate training overhead and risk model misspecification under sample-scarce conditions where $N_{\text{samp}} \ll 100$.

To overcome the collective limitations identified in the existing literature, KDS-EL establishes a closed-loop, knowledge-guided evolutionary learning framework that unifies offline domain priors, online memory harvesting, and stage-aware objective scheduling. Specifically, normalized mutual information scores are embedded as continuous Gaussian-perturbed seeds, preserving the full search space while guiding the swarm toward an ultra-sparse initial basin, complemented by an adaptive re-initialization mechanism to recover exploratory diversity upon stagnation (Section 3.2). Throughout evolution, cross-iteration fitness feedback is harvested into a cumulative importance matrix $\mathbf{W} \in \mathbb{R}^D$ governed by exponential forgetting, enabling the KER operator to execute targeted, directional elite flips within a strictly bounded computational budget (Section 3.3). Finally, a dynamic Two-Phase Weight Scheduling strategy temporally decouples early coarse feature pruning from late high-precision accuracy refinement, achieving simultaneous gains in classification accuracy, feature compactness, and runtime efficiency (Section 3.4). Table 1 systematically summarizes the methodological taxonomy and structural comparison between KDS-EL and all eleven benchmark feature selection algorithms investigated in this study across seven orthogonal technical dimensions.

3. The Proposed Method

To address the severe challenges of ultra-high-dimensional feature selection ($D \gg 10^3, N_{\text{samp}} \ll 100$), this section presents the proposed KDS-EL framework in detail. Specifically, the macro data-flow architecture and algorithmic flowchart of KDS-EL are outlined in Section 3.1. The prior-guided sparse initialization and adaptive recovery strategies are detailed in Section 3.2. Section 3.3 analyzes the search failure of conventional random perturbations, and formulates the closed-loop mechanism of cumulative importance learning and knowledge-guided elite refinement. The dynamic two-phase weight scheduling strategy is presented in Section 3.4. Finally, the overall algorithmic procedure and its computational complexity analysis are synthesized in Section 3.5.

3.1. The framework of KDS-EL

The macro data-flow architecture and problem mapping of KDS-EL for high-dimensional feature selection are systematically illustrated in Fig. 1. Given a high-dimensional dataset $\mathcal{D}$, the search pipeline begins by extracting filter-level mutual information to construct a continuous prior relevance landscape. An initial swarm of continuous positions is generated via hybrid prior seeding and deterministically mapped into sparse binary candidate masks. Throughout optimization, these candidate subsets are transferred to the KDS-EL optimization engine, which coordinates collaborative dual-swarm co-evolution, dynamic two-phase fitness evaluations, cumulative feature-importance harvesting, and targeted elite refinement in closed-loop synergy with a wrapper classification evaluation engine. Upon meeting the convergence criterion, the framework outputs the optimal compact biomarker subset $\mathbf{x}^\star$ alongside its validated classification performance.

The execution logic and operational flowchart across evolutionary generations are depicted in Fig. 2. In KDS-EL, a swarm of N candidate solutions across D features is formally represented as an $N \times D$ binary decision matrix $\mathbf{X}$:

$$\mathbf{X} = \begin{bmatrix} \mathbf{x}_1 \\ \mathbf{x}_2 \\ \vdots \\ \mathbf{x}_N \end{bmatrix} = \begin{bmatrix} x_{1,1} & x_{1,2} & \cdots & x_{1,D} \\ x_{2,1} & x_{2,2} & \cdots & x_{2,D} \\ \vdots & \vdots & \ddots & \vdots \\ x_{N,1} & x_{N,2} & \cdots & x_{N,D} \end{bmatrix} \in \{0, 1\}^{N \times D}, \tag{9}$$

where the i-th row vector $\mathbf{x}_i = [x_{i,1}, x_{i,2}, \ldots, x_{i,D}] \in \{0, 1\}^D$ denotes an individual candidate solution, with $x_{i,j} = 1$ indicating that feature f_j is selected and $x_{i,j} = 0$ indicating exclusion.

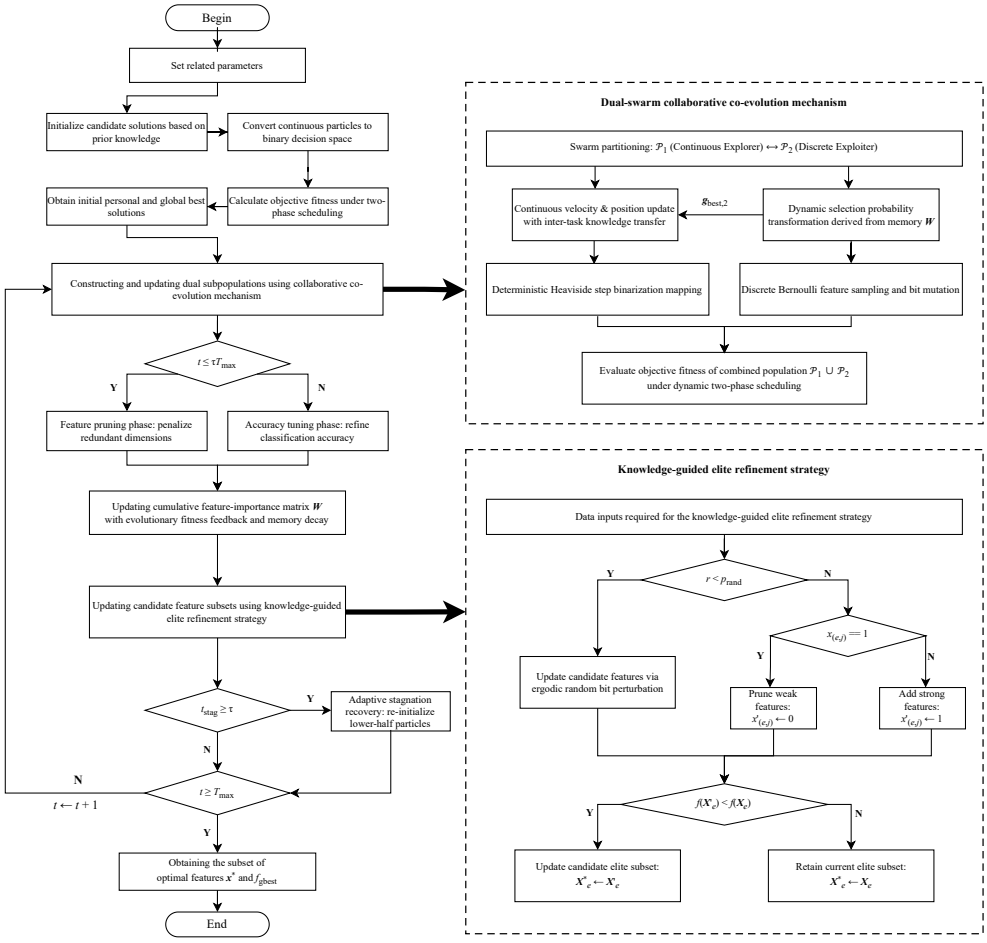


Figure 2: Overall flowchart for feature selection via KDS-EL.

To balance global exploration and local exploitation in sparse high-dimensional spaces, the population matrix $\mathbf{X}$ is partitioned into two collaborative subpopulations $\mathcal{P}_1$ and $\mathcal{P}_2$ of equal size. Subpopulation $\mathcal{P}_1$ explores continuous coordinates via velocity–position updates with an inter-task knowledge transfer channel guided by the global best solution of $\mathcal{P}_2$. Concurrently, $\mathcal{P}_2$ conducts discrete probability-driven feature sampling informed by the accumulated feature-importance landscape. Compared with conventional single-population architectures, this dual-subpopulation design establishes complementary search dynamics: continuous swarm movement preserves momentum to explore unvisited spaces, while discrete sampling focuses evaluations on verified high-relevance dimensions, effectively mitigating premature stagnation.

During iterative optimization, KDS-EL coordinates three complementary mechanisms into a closed loop. First, population initialization injects Gaussian-perturbed mutual information priors to generate sparse candidate masks, while an adaptive re-initialization mechanism recovers population diversity whenever search progress halts. Second, differential fitness improvements across generations are accumulated into a continuous importance matrix under exponential memory decay, which actively guides the KER operator to perform targeted directional bit flips under a greedy one-to-one in-place replacement rule. Third, the dynamic TPS strategy temporally shifts objective weights across evolutionary stages, enforcing aggressive feature pruning in the early phase and fine-grained accuracy tuning in the late phase. By continuously cycling through these coordinated stages, KDS-EL systematically prunes non-informative dimensions while progressively refining compact biomarker signatures.

3.2. Population initialization and adaptive recovery

In ultra-high-dimensional feature selection, standard uniform random initialization activates each feature with equal probability, generating excessively dense candidate subsets with approximately half the features active ($d \approx D/2$). Evaluating these dense candidate subsets exhausts computational budgets on non-informative dimensions and traps search trajectories in noisy local basins. To establish an informative yet diverse initial population while mitigating mid-evolution search stagnation, KDS-EL introduces a filter-wrapper hybrid seeding strategy coupled with an adaptive stagnation recovery mechanism, as illustrated in Fig. 3.

3.2.1. Prior-guided sparse population initialization

Given the training dataset instance $\mathcal{D} = \{(\mathbf{f}_i, y_i)\}_{i=1}^{N_{\text{samp}}}$ with feature vectors $\mathbf{f}_i \in \mathbb{R}^D$ and discrete class labels $y_i \in \{1, \ldots, C\}$, the mutual information scores $\{I(f_j; Y)\}_{j=1}^{D}$ evaluated via (8) quantify the linear and nonlinear dependencies between individual candidate dimensions and target categories. To establish a standardized probability profile across heterogeneous datasets, these raw mutual information scores are normalized into a prior relevance vector $\mathbf{m} = [m_1, \ldots, m_D]^T \in [0, 1]^D$:

$$m_j = \frac{I(f_j; Y) - \min_{1 \le k \le D} I(f_k; Y)}{\max_{1 \le k \le D} I(f_k; Y) - \min_{1 \le k \le D} I(f_k; Y) + \epsilon}, \quad j = 1, \ldots, D, \tag{10}$$

where $\epsilon > 0$ is a small numerical constant preventing division by zero.

To guarantee strict methodological integrity and eliminate selection bias, the filter prior calculation is architecturally restricted to the training partition presented to the optimizer. In an inductive machine learning deployment where an external unseen test partition $\mathcal{D}_{\text{test}}$ or an outer nested evaluation loop is prescribed, the prior relevance vector $\mathbf{m}$ and the subsequent evolutionary wrapper search are executed strictly on the training partition $\mathcal{D}_{\text{train}}$, ensuring that unseen validation samples remain completely unobserved throughout all search stages. In the standardized benchmark optimization protocol adopted in this work (Section 4.1), $\mathcal{D}$ serves as the complete benchmark problem instance, within which an internal 5-fold cross-validation scheme functions as the wrapper fitness evaluation surrogate.

To prevent population homogenization while guiding search trajectories toward informative dimensions, candidate particles are initialized through a hybrid seeding scheme parameterized by a seeding ratio $\rho \in (0, 1)$. Continuous

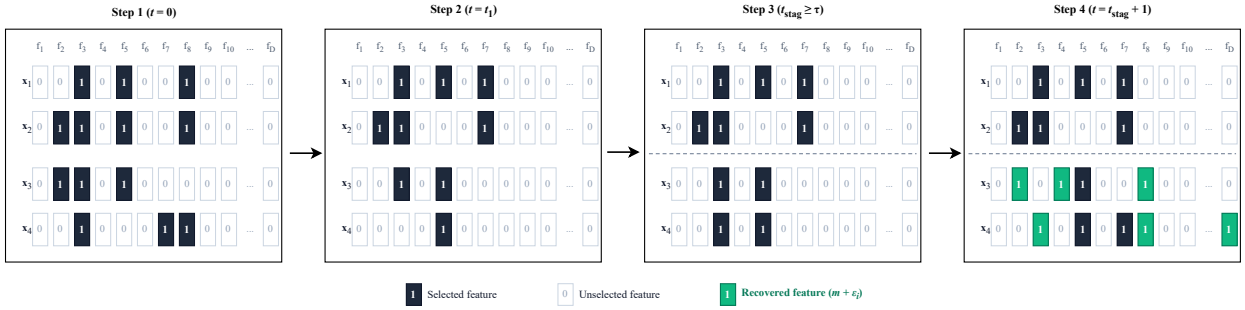


Figure 3: Process of prior-guided sparse initialization and adaptive recovery.

position vectors $\mathbf{p}_i = [p_{i,1}, \ldots, p_{i,D}] \in [0,1]^D$ are constructed as:

$$\mathbf{p}_i = \begin{cases} \text{clip}\left(\mathbf{m} + \boldsymbol{\varepsilon}_i,\ 0,\ 1\right), & 1 \leq i \leq \lfloor \rho N \rfloor, \\ \mathbf{u}_i, & \lfloor \rho N \rfloor < i \leq N, \end{cases} \tag{11}$$

where $\boldsymbol{\varepsilon}_i \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}_D)$ represents zero-mean Gaussian perturbation with standard deviation σ, $\mathbf{u}_i \sim \mathcal{U}(0,1)^D$ provides uniform stochastic coverage across unguided dimensions, and $\text{clip}(z, 0, 1) = \max(0, \min(1, z))$ restricts continuous coordinates to the unit hypercube.

Continuous position vectors $\mathbf{p}_i$ are subsequently transformed into binary decision vectors $\mathbf{x}_i = [x_{i,1}, \ldots, x_{i,D}] \in \{0,1\}^D$ via a deterministic Heaviside step mapping with threshold $\theta \in (0.5, 1)$:

$$x_{i,j} = \begin{cases} 1, & \text{if } p_{i,j} \geq \theta, \\ 0, & \text{otherwise}, \end{cases} \qquad j = 1, \ldots, D. \tag{12}$$

By setting the cut-off threshold $\theta > 0.50$, only features possessing above-average prior relevance are initialized in active states. This produces an intrinsically sparse initial population that biases exploration toward high-potential biomarker candidates, thereby resolving the cold-start bottleneck, reducing early evaluation latency, and accelerating convergence toward promising feature subspaces.

3.2.2. Adaptive stagnation recovery

In multi-modal high-dimensional search spaces, population diversity inevitably decays across generations, which may lead the swarm into suboptimal attractors. To resolve this issue, KDS-EL incorporates an adaptive re-initialization routine triggered whenever the global-best fitness fails to improve for τ consecutive generations.

As depicted in Fig. 3, particles are first sorted in ascending order of their personal-best fitness values. The superior half of the swarm is retained intact to safeguard historical search trajectories and verified biomarker combinations. Concurrently, the lower-performing half is refreshed by re-sampling continuous positions around the prior relevance vector $\mathbf{m}$ under Gaussian perturbation:

$$\mathbf{p}_i \leftarrow \text{clip}\left(\mathbf{m} + \boldsymbol{\varepsilon}_i,\ 0,\ 1\right), \quad x_{i,j} \leftarrow \mathbb{I}(p_{i,j} \geq \theta), \quad \forall j = 1, \ldots, D, \tag{13}$$

where $\mathbb{I}(\cdot)$ denotes the indicator function. The refreshed particles are immediately re-evaluated under current objective weights, and the stagnation counter is reset. By combining elite preservation with targeted prior re-injection, this mechanism restores exploratory diversity within high-potential subspaces without discarding accumulated optimization experience.

3.3. Cumulative importance learning and elite refinement

To overcome the search amnesia and destructive perturbation bottlenecks of conventional meta-heuristics, KDS-EL establishes a closed-loop knowledge learning mechanism comprising online importance accumulation and targeted directional elite refinement.

In ultra-high-dimensional feature selection ($D \gg 10^3$), traditional evolutionary wrappers rely on undirected stochastic bit flips with a uniform mutation probability $p_{\text{mut}} = 1/D$. Under this blind perturbation regime, meta-heuristics inevitably suffer from two fatal structural search lesions. First, when an established, highly informative biomarker ($x_{e,j} = 1$) is blindly mutated to 0, the delicate multi-feature synergy is abruptly dismantled, inducing severe classification degradation. Second, when an uninformative background noise feature ($x_{e,j} = 0$) is randomly flipped to 1, the candidate subset expands unnecessarily, escalating distance evaluation latency and precipitating classifier overfitting. Because the probability of hitting the few decisive dimensions within a 2^D combinatorial neighborhood is infinitesimal ($\mathcal{O}(1/D) \approx 10^{-4}$), undirected mutation squanders the vast majority of function evaluations on destructive trials. Furthermore, conventional meta-heuristics exhibit severe search amnesia, as historical fitness feedback from intermediate iterations is discarded immediately after particle position updates.

3.3.1. Online feature importance learning

To systematically harvest search experience across generations, KDS-EL maintains a continuous cumulative feature-importance vector $\mathbf{W} = [W_1, \ldots, W_D]^T \in \mathbb{R}^D$, initialized to $\mathbf{0}$.

At generation t, whenever a candidate particle $\mathbf{x}_i^{(t)}$ achieves a strict fitness improvement over its historical personal best $\mathbf{p}_{\text{best},i}$, the differential feature set $C_i^{(t)}$ comprising features whose selection status changed is extracted:

$$C_i^{(t)} = \left\{ j \in \{1, \ldots, D\} \;\middle|\; x_{i,j}^{(t)} \neq p_{\text{best},i,j} \right\}. \tag{14}$$

Let $\delta_i^{(t)} = f(\mathbf{p}_{\text{best},i}, \mathbf{w}(t)) - f(\mathbf{x}_i^{(t)}, \mathbf{w}(t)) > 0$ denote the scalar fitness improvement. Each altered feature $j \in C_i^{(t)}$ receives a directional reward proportional to the magnitude of $\delta_i^{(t)}$:

$$W_j \leftarrow W_j + \delta_i^{(t)} \cdot (2x_{i,j}^{(t)} - 1), \quad \forall j \in C_i^{(t)}, \tag{15}$$

where the term $(2x_{i,j}^{(t)} - 1)$ yields $+1$ if feature j was newly activated and -1 if deactivated. Candidate solutions that fail to improve fitness ($\delta_i^{(t)} \leq 0$) leave $\mathbf{W}$ unchanged.

To prevent early-discovered features from dominating the importance landscape indefinitely and maintain sensitivity toward emergent synergistic biomarker interactions, an exponential memory decay is applied at the conclusion of each generation:

$$\mathbf{W} \leftarrow \gamma \cdot \mathbf{W}, \tag{16}$$

where $\gamma \in (0,1)$ is the memory retention factor. Consequently, features that consistently contribute to fitness improvements sustain high positive scores, whereas transiently selected or uninformative dimensions exponentially fade toward zero.

3.3.2. Knowledge-guided elite refinement

Elite individuals represent the most promising feature subsets discovered by the swarm; however, they may still retain redundant dimensions or lack subtle synergistic biomarkers. Rather than blind perturbation, the KER operator exploits the continuous importance landscape $\mathbf{W}$ to guide local mutation toward high-potential dimensions under bounded computation ($k_e \ll N$).

At each generation, the top k_e individuals ranked by fitness are selected into the elite pool. For each elite candidate $\mathbf{x}_e \in \{0,1\}^D$, the number of perturbed features is set to $k_{\text{flip}} = \max(1, \lfloor \phi D \rfloor)$, where $\phi \in (0,1)$ represents the perturbation fraction. As depicted in the data-flow diagram in Fig. 4, KER probabilistically executes one of two operational pathways:

Branch 1: Ergodic random exploration (p_{rand}): To preserve search ergodicity and prevent the elite pool from stagnating in deceptive local optima, a feature index subset $\mathcal{S}_{\text{rand}} \subset \{1, \ldots, D\}$ of size k_{flip} is sampled uniformly at random, and the corresponding bits are inverted:

$$x'_{e,j} \leftarrow 1 - x_{e,j}, \quad \forall j \in \mathcal{S}_{\text{rand}}, \quad |\mathcal{S}_{\text{rand}}| = k_{\text{flip}}. \tag{17}$$

Branch 2: Knowledge-guided directional refinement ($1 - p_{\text{rand}}$): For each feature $j \in \{1, \ldots, D\}$, a state-dependent directional flip score σ_j is computed:

$$\sigma_j = (1 - 2x_{e,j}) \cdot W_j = \begin{cases} +W_j, & \text{if } x_{e,j} = 0, \\ -W_j, & \text{if } x_{e,j} = 1. \end{cases} \tag{18}$$

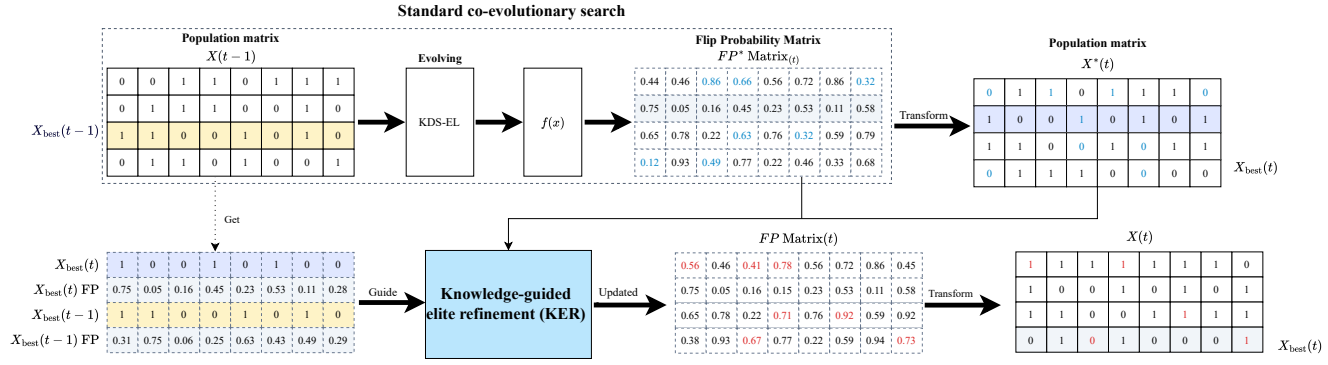


Figure 4: Process of the knowledge-guided elite refinement strategy.

Algorithm 1 Knowledge-guided elite refinement

Require: Elite candidate solution $\mathbf{x}_e \in \{0,1\}^D$, cumulative importance vector $\mathbf{W} \in \mathbb{R}^D$, perturbation fraction $\phi \in (0,1)$, exploration probability $p_{\text{rand}} \in [0,1]$, dynamic weight vector $\mathbf{w}(t) \in [0,1]^2$
Ensure: Refined elite candidate solution $\mathbf{x}_e^\star \in \{0,1\}^D$

$k_{\text{flip}} \leftarrow \max(1, \lfloor \phi \cdot D \rfloor)$
$\mathbf{x}'_e \leftarrow \mathbf{x}_e$ ▷ Initialize mutant candidate
Sample random variable $r \sim \mathcal{U}(0,1)$
if $r < p_{\text{rand}}$ **then** ▷ Branch 1: Ergodic random exploration
  Sample random feature index subset $\mathcal{S}_{\text{rand}} \subset \{1,\dots,D\}$ with $|\mathcal{S}_{\text{rand}}| = k_{\text{flip}}$
  $x'_{e,j} \leftarrow 1 - x_{e,j}, \quad \forall j \in \mathcal{S}_{\text{rand}}$
else ▷ Branch 2: Knowledge-guided directional refinement
  for $j = 1$ **to** D **do**
    $\sigma_j \leftarrow (1 - 2x_{e,j}) \cdot W_j$ ▷ Compute directional flip score via (18)
  end for
  Identify index subset $\mathcal{S}_{\text{know}}$ of size k_{flip} with top σ_j scores via (19)
  $x'_{e,j} \leftarrow 1 - x_{e,j}, \quad \forall j \in \mathcal{S}_{\text{know}}$
end if
Evaluate candidate fitness $f(\mathbf{x}'_e, \mathbf{w}(t))$ via (22)
if $f(\mathbf{x}'_e, \mathbf{w}(t)) < f(\mathbf{x}_e, \mathbf{w}(t))$ **then** ▷ Greedy 1-to-1 in-place replacement
  $\mathbf{x}_e^\star \leftarrow \mathbf{x}'_e$
else
  $\mathbf{x}_e^\star \leftarrow \mathbf{x}_e$
end if
return $\mathbf{x}_e^\star$

A positive flip score $\sigma_j > 0$ indicates either an unselected feature ($x_{e,j} = 0$) with positive historical contribution ($W_j > 0$), or an active feature ($x_{e,j} = 1$) whose deactivation has historically improved fitness ($W_j < 0$). The feature subset $\mathcal{S}_{\text{know}}$ containing the k_{flip} features with the largest directional flip scores is identified:

$$\mathcal{S}_{\text{know}} = \arg \max_{\substack{\mathcal{S} \subseteq \{1,\dots,D\} \\ |\mathcal{S}| = k_{\text{flip}}}} \sum_{j \in \mathcal{S}} \sigma_j, \tag{19}$$

and their binary states are inverted: $x'_{e,j} \leftarrow 1 - x_{e,j}, \forall j \in \mathcal{S}_{\text{know}}$.

Greedy one-to-one in-place replacement: The perturbed candidate $\mathbf{x}'_e$ is evaluated using the objective function $f(\mathbf{x}'_e, \mathbf{w}(t))$. A greedy acceptance rule determines solution admission:

$$\mathbf{x}_e^\star = \begin{cases} \mathbf{x}'_e, & \text{if } f(\mathbf{x}'_e, \mathbf{w}(t)) < f(\mathbf{x}_e, \mathbf{w}(t)), \\ \mathbf{x}_e, & \text{otherwise.} \end{cases} \tag{20}$$

Upon achieving a strict fitness improvement, $\mathbf{x}'_e$ immediately replaces $\mathbf{x}_e$ in the population and updates the global-best solution if $f(\mathbf{x}'_e, \mathbf{w}(t)) < f_{\text{gbest}}$. Because the elite pool size k_e is fixed ($k_e \ll N$), KER operates within an exact bounded overhead of k_e evaluations per generation. The complete procedural logic of KER is formalized in Algorithm 1.

3.4. Dynamic two-phase weight scheduling

In high-dimensional feature selection, wrapper optimization must balance two fundamentally conflicting objectives: minimizing classification error and minimizing the selected feature subset cardinality. In the early evolutionary phase, the search space is dominated by an overwhelming number of redundant and noisy dimensions; thus, the primary optimization imperative is aggressive dimensionality pruning. In contrast, during the late evolutionary phase when search trajectories have converged to a compact subspace, the priority shifts to fine-tuning synergistic biomarker interactions to maximize classification accuracy. Conventional static weighting schemes with fixed trade-off coefficients fail to reconcile these temporally distinct demands, frequently inducing either inadequate early feature elimination or premature penalization of critical predictive features in later stages.

To resolve this temporal conflict, KDS-EL introduces the dynamic TPS strategy, which decouples the evolutionary horizon into two distinct functional phases. The total generation budget T is divided at a split point parameterized by the phase transition ratio $\kappa \in (0,1)$. The dynamic objective weight vector $\mathbf{w}(t) = [w_a(t), w_f(t)] \in [0,1]^2$, satisfying $w_a(t) + w_f(t) = 1$, is formulated as:

$$\mathbf{w}(t) = \begin{cases} [w_{a,1}, w_{f,1}], & 1 \le t \le \lfloor \kappa T \rfloor, \\ [w_{a,2}, w_{f,2}], & \lfloor \kappa T \rfloor < t \le T, \end{cases} \tag{21}$$

where the weight coefficients satisfy the ordering $w_{f,1} > w_{f,2}$ and $w_{a,1} < w_{a,2}$.

At generation t, the unified scalar fitness $f(\mathbf{x}, \mathbf{w}(t))$ of a binary feature subset $\mathbf{x} \in \{0,1\}^D$ is evaluated as:

$$f(\mathbf{x}, \mathbf{w}(t)) = w_a(t) \cdot (1 - \text{Acc}(\mathbf{x})) + w_f(t) \cdot \left(\frac{\|\mathbf{x}\|_0}{D} \right), \tag{22}$$

where $\text{Acc}(\mathbf{x}) \in [0,1]$ represents the 5-fold cross-validation classification accuracy evaluated by a k-NN classifier, and $\|\mathbf{x}\|_0 = \sum_{j=1}^{D} x_j$ denotes the ℓ_0-norm quantifying the number of active features.

During Phase 1 ($1 \le t \le \lfloor \kappa T \rfloor$), the elevated feature penalty $w_{f,1} = 0.30$ enforces severe selective pressure against non-informative features, rapidly compressing the active candidate dimensionality to a compact fraction of D. Subsequently, upon entering Phase 2 ($\lfloor \kappa T \rfloor < t \le T$), the weight vector transitions to prioritize classification accuracy ($w_{a,2} = 0.95$), empowering the swarm to explore subtle multi-gene synergies over the condensed feature subspace without being prematurely penalized for retaining essential predictive biomarkers.

3.5. Overall algorithmic procedure and complexity analysis

Algorithm 2 synthesizes the overall execution pipeline of KDS-EL into a structured multi-stage optimization framework. The complete workflow coordinates offline filter relevance extraction, knowledge-informed hybrid seeding, dynamic two-phase objective scheduling, collaborative dual-subpopulation search, differential importance harvesting, and targeted elite refinement.

3.5.1. Computational complexity analysis

The total computational overhead of KDS-EL is decoupled into two distinct phases: offline preprocessing and online generational evolution.

Offline preprocessing complexity: Evaluating mutual information scores across D features with N_{samp} training samples requires $O(D \cdot N_{\text{samp}})$ operations. Normalizing and sorting these scores incurs $O(D \log D)$ time. Because this procedure is executed exactly once prior to evolutionary search, the offline preprocessing complexity is $\mathcal{T}_{\text{pre}} = O(D \cdot N_{\text{samp}} + D \log D)$, which represents a negligible computational fraction when $N_{\text{samp}} \ll 100$.

Online generational complexity: In each generation $t \in \{1,\dots,T\}$, the time complexity comprises three components:

(1) *Subpopulation updates and knowledge transfer*: Continuous particle updates in $\mathcal{P}_1$ and discrete probability sampling in $\mathcal{P}_2$ both require $O(N \cdot D)$ arithmetic operations per generation.

Algorithm 2 The overall framework of KDS-EL

Require: Training dataset $\mathcal{D} = \{(\mathbf{f}_i, y_i)\}_{i=1}^{N_{\text{samp}}}$, population size $N \in \mathbb{Z}_+$, maximum generations $T \in \mathbb{Z}_+$, binarization threshold $\theta \in (0,1)$, prior seeding ratio $\rho \in (0,1)$, perturbation scale $\sigma \in \mathbb{R}_+$, forgetting factor $\gamma \in (0,1)$, elite pool size $k_e \in \mathbb{Z}_+$, perturbation fraction $\phi \in (0,1)$, random branch probability $p_{\text{rand}} \in [0,1]$, phase transition split ratio $\kappa \in (0,1)$, stagnation threshold $\tau \in \mathbb{Z}_+$

Ensure: Optimal compact feature subset $\mathbf{x}^\star \in \{0,1\}^D$ and optimal fitness $f_{\text{gbest}} \in \mathbb{R}$

Stage 1: Filter-level relevance extraction

Compute normalized mutual information vector $\mathbf{m} \in [0,1]^D$ via (8) and (10)

Stage 2: Knowledge-informed hybrid population seeding

Initialize population $\mathcal{P} = \{\mathbf{x}_i\}_{i=1}^N$ with $\lfloor \rho N \rfloor$ Gaussian MI seeds and $(N - \lfloor \rho N \rfloor)$ uniform particles via (11)–(12)

Initialize importance vector $\mathbf{W} \leftarrow \mathbf{0} \in \mathbb{R}^D$, stagnation counter $c_{\text{stall}} \leftarrow 0$, and initial weights $\mathbf{w}(1) = [w_{a,1}, w_{f,1}]$ via (21)

Evaluate initial candidate fitness values $\{f(\mathbf{x}_i, \mathbf{w}(1))\}_{i=1}^N$ via (22); initialize personal bests $\{\mathbf{p}_{\text{best},i}\}_{i=1}^N$ and global best $\mathbf{g}_{\text{best}}$

Stage 3: Dynamic two-phase co-evolutionary loop

for $t = 1$ **to** T **do**

  Update dynamic objective weight vector $\mathbf{w}(t)$ according to (21)

  Update continuous Subpopulation $\mathcal{P}_1$ via continuous PSO with inter-task knowledge transfer [12]

  Update discrete Subpopulation $\mathcal{P}_2$ via discrete probability sampling guided by $\mathbf{W}$

  Evaluate candidate solution fitness $\{f(\mathbf{x}_i, \mathbf{w}(t))\}_{i=1}^N$ using 5-fold cross-validation k-NN classifier via (22)

  Update cumulative importance vector $\mathbf{W}$ with improving differentials and apply memory decay γ via (15)–(16)

  Update personal bests $\{\mathbf{p}_{\text{best},i}\}_{i=1}^N$ and identify top k_e elite candidate indices $\mathcal{E}$

  for each $e \in \mathcal{E}$ **do** ▷ Knowledge-guided elite refinement

    $\mathbf{p}_{\text{best},e} \leftarrow \text{KER}(\mathbf{p}_{\text{best},e}, \mathbf{W}, \phi, p_{\text{rand}}, \mathbf{w}(t))$ ▷ Invoke Algorithm 1

  end for

  if $\min_i f(\mathbf{p}_{\text{best},i}, \mathbf{w}(t)) < f(\mathbf{g}_{\text{best}}, \mathbf{w}(t))$ **then**

    Update global best $\mathbf{g}_{\text{best}} \leftarrow \arg\min_i f(\mathbf{p}_{\text{best},i}, \mathbf{w}(t))$ and reset $c_{\text{stall}} \leftarrow 0$

  else

    Increment stagnation counter $c_{\text{stall}} \leftarrow c_{\text{stall}} + 1$

  end if

  if $c_{\text{stall}} \geq \tau$ **then** ▷ Adaptive stagnation recovery

    Refresh lower-performing half of population using Gaussian-perturbed MI seeds via (13)

    Re-evaluate refreshed particles and reset $c_{\text{stall}} \leftarrow 0$

  end if

end for

return $\mathbf{x}^\star \leftarrow \mathbf{g}_{\text{best}}$, $f_{\text{gbest}} \leftarrow f(\mathbf{g}_{\text{best}}, \mathbf{w}(T))$

(2) *Classifier wrapper evaluation*: Evaluating candidate feature subsets dominates online computation. For a candidate solution $\mathbf{x}_i$ with active feature count $d_i = \|\mathbf{x}_i\|_0$, computing pairwise Euclidean distances under 5-fold cross-validation with a k-NN classifier incurs $O(N_{\text{samp}}^2 \cdot d_i)$ time. Across the swarm of size N, the total generational evaluation cost is $O(N \cdot N_{\text{samp}}^2 \cdot \bar{d}^{(t)})$, where $\bar{d}^{(t)} = \frac{1}{N}\sum_{i=1}^N d_i$ denotes the average active feature count at generation t.

(3) *Cumulative importance learning and elite refinement*: Extracting differential features and updating $\mathbf{W}$ requires $O(N \cdot D)$ operations. In KER, computing directional flip scores across D dimensions and identifying the top k_{flip} features takes $O(D \log D)$ time using partial quick-sort. Evaluating the k_e mutant candidates incurs $O(k_e \cdot N_{\text{samp}}^2 \cdot d_{\text{elite}})$ operations. Because $k_e \ll N$ ($k_e = 3$), this overhead remains strictly bounded.

Combining these components across T generations yields the total asymptotic time complexity of KDS-EL:

$$\mathcal{T}_{\text{total}} = O\left(T \cdot N \cdot N_{\text{samp}}^2 \cdot \bar{d}_{\text{selected}} + T \cdot N \cdot D\right), \tag{23}$$

where $\bar{d}_{\text{selected}}$ denotes the average active feature count across all evolutionary generations.

Asymptotic vs. wall-clock speedup: While KDS-EL exhibits the same asymptotic Big-O order as standard PSO-based feature selectors ($O(TNN_{\text{samp}}^2 D)$), it achieves an empirical wall-clock speedup exceeding 3.4× in practice. This dramatic efficiency gain originates directly from the prior-guided sparse initialization and dynamic TPS strategy: rather than evaluating dense subsets ($\bar{d} \approx D/2 \approx 5{,}000+$), KDS-EL aggressively compresses candidate dimensionality to $\bar{d}_{\text{KDS-EL}} \approx 80 \sim 150$ active features starting from the earliest generations. Because the k-NN distance computation cost scales strictly linearly with the number of active features d, compressing the active dimension by two orders of magnitude reduces the actual floating-point operation volume per evaluation by over 95%.

Spatial complexity: The memory consumption of KDS-EL comprises the population matrix $\mathbf{X} \in \{0,1\}^{N \times D}$, historical personal bests $\mathbf{P}_{\text{best}} \in \{0,1\}^{N \times D}$, the continuous importance vector $\mathbf{W} \in \mathbb{R}^D$, and the training dataset $\mathcal{D} \in \mathbb{R}^{N_{\text{samp}} \times D}$. The total spatial complexity is $\mathcal{S}_{\text{total}} = O\left((2N + N_{\text{samp}} + 1)D\right) = O\left((N + N_{\text{samp}})D\right)$, which scales strictly linearly with feature dimensionality D and fits comfortably within standard workstation memory even when $D > 10^5$.

4. Experimental results and discussion

This section outlines the experimental framework and presents a comprehensive empirical evaluation of KDS-EL across eight analytical dimensions: classification accuracy, feature subset compactness, fitness convergence dynamics, statistical significance, performance progression before and after algorithmic improvement, computational efficiency, hyperparameter sensitivity, and in-depth discussion on biomedical ceiling effects. All aggregate numbers are computed over the 20 benchmark datasets described in Section 4.1.

4.1. Experimental setup

This subsection outlines the experimental framework designed to evaluate the classification performance, dimensionality reduction capability, computational efficiency, and hyperparameter robustness of KDS-EL.

4.1.1. High-dimensional benchmark datasets

To validate the performance and generalization ability of KDS-EL across diverse problem domains, a total of 20 benchmark datasets widely adopted in the evolutionary feature selection literature [8, 11, 12] are selected. These datasets encompass diverse categories and sample sizes across biomedical gene-expression microarrays, text categorization corpora, image recognition, speech processing, and synthetic benchmarks. Such diverse data distributions and dimensional complexities effectively evaluate the generalization capability and feature reduction efficacy of feature selection algorithms under challenging conditions. The detailed characteristics of the 20 benchmark datasets are summarized in Table 2.

4.1.2. Comparison algorithms

We select two primary categories of comparison algorithms for analysis with the proposed KDS-EL framework, as summarized in Table 1. The first category comprises specialized single-objective evolutionary algorithms in the field of high-dimensional feature selection, which includes: FWPSO [8], PSO-CSM [11], SaWDE [9], IGPSO [10], and BPSO [7]. The relevant parameter settings are shown in Table 3.

Another category of representative algorithms encompasses multi-objective and multi-task feature selection approaches, including NSGA-II-FS [13], MOEA/D-FS [14], MOPSO-FS [15], fMOPSO-FS [16], W-MOPSO-FS [17], and MEL [12]. In addition, two constituent variants of KDS-EL are investigated in Section 4.2.5 to evaluate individual component contributions. Together with foundational baselines, this 11-algorithm comparison suite systematically spans classical search paradigms and state-of-the-art developments published between 2022 and 2025, ensuring a balanced, rigorous, and reproducible empirical evaluation.

4.1.3. Evaluation metrics and statistical test protocols

Algorithm performance is evaluated across three primary quantitative criteria: mean classification accuracy alongside standard deviation, the number of selected features alongside the feature selection ratio (FSR = $\frac{\bar{d}}{D} \times 100\%$), and computational runtime in seconds. The k-nearest neighbor classifier with $k = 3$ serves as the base evaluation model due to its non-parametric nature, parameter-free reproducibility, and high sensitivity to candidate feature subset discriminability. Following the standard benchmarking protocol established in high-dimensional evolutionary feature selection literature [8, 11, 12], candidate feature subsets are evaluated using 5-fold stratified cross-validation as the internal wrapper fitness surrogate across all problem instances. Under this standardized formulation, each benchmark dataset $\mathcal{D}$ is treated as a self-contained combinatorial optimization instance; the 5-fold cross-validation classification accuracy guides the evolutionary search trajectory and evaluates the discriminative quality of the optimal feature subsets under identical computational budgets across all competing methods. To guarantee a rigorous and methodologically equitable comparison between single-objective wrappers and multi-objective evolutionary algorithms that yield an entire Pareto non-dominated front $\mathcal{PF}$, a standardized post-Pareto decision-making protocol is enforced. In accordance with established multi-objective feature selection benchmarks [16, 17], the final operational solution $\mathbf{x}^\star \in \mathcal{PF}$ is uniquely extracted as the non-dominated

candidate that achieves the minimal classification error on the cross-validation folds, with ties broken by smaller subset cardinality:

$$\mathbf{x}^{\star} = \arg\min_{\mathbf{x}\in\mathcal{P}\mathcal{F}} \Big(\text{ErrorRate}(\mathbf{x}),\ \|\mathbf{x}\|_0\Big). \tag{24}$$

This operational protocol ensures that all competing paradigms are systematically evaluated at their maximum predictive accuracy.

To guarantee experimental fairness and statistical reliability, every algorithm is executed with a population size of $N = 20$ candidate solutions over $T_{\max} = 100$ generations across 10 independent runs with distinct random seeds. In ultra-high-dimensional wrapper feature selection, each fitness evaluation requires training and validating a classifier across thousands or tens of thousands of candidate dimensions (e.g., up to 54,675 dimensions in *T1D*), imposing an exceptionally intensive computational burden of $N \times T_{\max} = 2{,}000$ cross-validation evaluations per single run. Executing 10 independent runs per dataset corresponds to 50 distinct stratified cross-validation folds, which provides an adequate sample size to capture classifier generalization variance and algorithmic stability while remaining computationally tractable across an extensive 14-algorithm comparison suite. This 10-run protocol strictly aligns with the benchmark evaluation standard established in recent top-tier literature on ultra-high-dimensional feature selection, most notably the multi-task evolutionary learning framework MEL published in *IEEE Transactions on Knowledge and Data Engineering* [12], thereby guaranteeing an equitable and reproducible baseline comparison.

Furthermore, to determine whether the empirical advantages of KDS-EL over competing paradigms are statistically significant rather than stochastic artifacts, we strictly follow the multi-problem non-parametric statistical testing framework recommended by Demšar [39]. In this protocol, statistical power is governed by the number of independent benchmark problems ($N_{\text{datasets}} = 20$), with each problem represented by the paired summary performance across the independent runs. Specifically, the non-parametric Friedman test [40] is utilized to establish global omnibus algorithm rankings across all 20 benchmark datasets. Pairwise Wilcoxon signed-rank tests at the $\alpha = 0.05$ significance level assess directional superiority between KDS-EL and each competitor, with the Holm–Bonferroni step-down procedure [41] applied to rigorously control the family-wise error rate across multiple hypothesis comparisons.

Table 2: Characteristics of the 20 benchmark datasets.

No.	Dataset	Samples (n)	Features (D)	Classes (C)	Domain
1	Adenoma	36	7,457	2	Gene expression
2	ALL_AML	72	7,129	2	Gene expression
3	ALL1	128	12,625	2	Gene expression
4	ALL4	93	12,625	2	Gene expression
5	Colon	62	2,000	2	Gene expression
6	DLBCL	77	7,129	2	Gene expression
7	Gastric	65	22,645	2	Gene expression
8	Gastric1	144	22,283	2	Gene expression
9	Gastric2	124	22,283	2	Gene expression
10	Leukaemia	72	7,129	2	Gene expression
11	Lymphoma	45	4,026	2	Gene expression
12	Myeloma	173	12,625	2	Gene expression
13	Prostate	102	12,625	2	Gene expression
14	SRBCT	83	2,308	4	Gene expression
15	T1D	101	54,675	2	Gene expression
16	BASEHOCK	1,993	4,862	2	Text
17	PCMAC	1,943	3,289	2	Text
18	COIL20	1,440	1,024	20	Image
19	ISOLET	1,560	617	26	Speech
20	madelon	2,600	500	2	Synthetic

4.2. Experimental analysis

This subsection presents a comprehensive empirical evaluation of KDS-EL across eight complementary analytical dimensions: classification accuracy, feature subset compactness, fitness convergence dynamics, statistical significance, performance progression before and after algorithmic improvement, computational efficiency, hyperparameter sensitivity, and in-depth discussion on biomedical ceiling effects.

Table 3: Relevant parameter settings for all algorithms.

Algorithm / Component	Parameter setting
Common experimental parameters:	
Population size N	20
Maximum generations $T_{\max}$	100
Classifier	3-NN with 5-fold stratified CV
Binarization threshold θ	0.60
Independent runs	10
Proposed KDS-EL framework:	
Subpopulations	$N_1 = 10,\ N_2 = 10$
Inertia weight ω	0.90
Acceleration coefficients c_1, c_2, c_3	2.0, 2.0, 2.0
Knowledge decay factor γ	0.99
Perturbation fraction ϕ	0.01
Elite pool size k_e	3
Random exploration rate p_{rand}	0.10
MI initialization ratio ρ	0.50
Stagnation threshold τ	20 generations
Phase split ratio κ	0.70
Phase-1 weights $\mathbf{w}_1$	[0.70, 0.30]
Phase-2 weights $\mathbf{w}_2$	[0.95, 0.05]
Ablative variants:	
KDS-EL w/o TPS	Static weights $\mathbf{w}_s = [0.90,\ 0.10]$
KDS-EL w/o TPS&KER	Without KER and TPS, $\mathbf{w}_s = [0.90,\ 0.10]$
Original MEL [12]	Uniform init, $\mathbf{w}_s = [0.90,\ 0.10]$
Competitor algorithms:	
FWPSO [8]	$\omega \in [0.4, 0.9],\ c_1 = c_2 = 2.0,\ \alpha_{\text{sa}} = 0.95$
PSO-CSM [11]	$\omega = 0.7298,\ c_1 = c_2 = 1.49618,\ p_m = 0.05$
SaWDE [9]	$F \in [0.1, 0.9],\ CR \in [0.1, 0.9]$
IGPSO [10]	$\omega \in [0.4, 0.9],\ c_1 = c_2 = 2.0$, IG scoring
BPSO [7]	$\omega \in [0.4, 0.9],\ c_1 = c_2 = 2.0,\ V_{\max} = 6.0$
NSGA-II-FS [13]	$p_c = 0.9,\ \eta_c = 20,\ p_m = 1/D,\ \eta_m = 20$
MOEA/D-FS [14]	Neighborhood $T_n = 5$, Tchebycheff
MOPSO-FS [15]	Archive 100, grid inflation $\alpha_g = 0.1$
fMOPSO-FS [16]	Archive 100, $\omega = 0.7298,\ c_{1,2} = 1.49618$
W-MOPSO-FS	Archive 100, dynamic filter–wrapper

4.2.1. Comparison of classification accuracy

The KDS-EL framework was tested with 13 comparison algorithms on 20 benchmark datasets with high-dimensional feature selection. Table 4 presents the results of the experiments for average classification accuracy and standard deviation, with the optimal values in bold.

First, comparing the KDS-EL algorithm to the 13 comparison algorithms overall, KDS-EL achieved the highest average classification accuracy of 0.9574 across the benchmark suite, securing the highest score on 15 of the 20 high-dimensional datasets. Among them, on ultra-high-dimensional microarrays with severe sample sparsity, KDS-EL delivers substantial performance advantages, reaching classification accuracies of 0.9522 on the 12,625-dimensional *ALL4*, 0.9500 on *Colon*, and 0.8598 on the 54,675-dimensional *T1D*, consistently establishing distinct margins over competing single- and multi-objective algorithms. Secondly, compared to established single-objective evolutionary algorithms such as FWPSO, PSO-CSM, SaWDE, IGPSO, and BPSO, KDS-EL improves overall mean accuracy from the 0.8698–0.9177 range to 0.9574, indicating that closed-loop knowledge transfer and Two-Phase Weight Scheduling effectively overcome the blind stochastic trials of standard swarm operators. Compared with multi-objective methods including W-MOPSO-FS, fMOPSO-FS, NSGA-II-FS, MOEA-D-FS, and MOPSO-FS, KDS-EL maintains a consistent advantage of over 0.04 to 0.05 in mean classification score. Furthermore, compared with the original MEL algorithm, KDS-EL achieves an accuracy improvement of 0.0706, which indicates that mutual information initialization and knowledge-guided elite refinement effectively enhance the algorithm's biomarker mining capability and local exploitation. Overall, the KDS-EL algorithm searches for a superior subset of features and obtains higher classification accuracy.

A closer examination of the individual datasets and algorithmic dispersion provides further insights into the search behavior and robustness of KDS-EL. On the few moderate-dimensional datasets where KDS-EL does not record the nominal highest accuracy, such as *BASEHOCK*, *COIL20*, and *ISOLET*, competitors such as NSGA-II-FS achieve marginally higher accuracy by 0.003 to 0.020 only by retaining massive subsets of 185 to 2,073 features, whereas KDS-EL attains competitive accuracy with only 60 to 134 features, achieving an

Table 4: Comparison of classification accuracy (mean ± std).

Dataset	KDS-EL	w/o TPS	w/o KER	MEL	FWPSO	PSO-CSM	SaWDE	IGPSO	BPSO	W-MPSO	fMPSO	NSGA-II	MOEA/D	MOPSO
ALL1	0.9998±0.0001	0.9996±0.0000^{+}	0.9996±0.0000^{+}	0.9664±0.0023^{+}	0.9952±0.0015^{+}	0.9900±0.0002^{+}	0.9710±0.0011^{+}	0.9802±0.0002^{+}	0.9534±0.0003^{+}	**1.0000±0.0000**$^{-}$	**1.0000±0.0000**$^{-}$	**1.0000±0.0000**$^{-}$	0.9985±0.0031$^{\approx}$	**1.0000±0.0000**$^{-}$
ALL4	**0.9522±0.0061**	0.9418±0.0113$^{\approx}$	0.9415±0.0112^{+}	0.8704±0.0661^{+}	0.8617±0.0304^{+}	0.8363±0.0135^{+}	0.8062±0.0217^{+}	0.7914±0.0179^{+}	0.7569±0.0069^{+}	0.8068±0.0047^{+}	0.8015±0.0089^{+}	0.7694±0.0118^{+}	0.7737±0.0184^{+}	0.7654±0.0074^{+}
ALL_AML	**0.9995±0.0002**	0.9990±0.0002^{+}	0.9886±0.0016^{+}	0.9539±0.0008^{+}	0.9850±0.0067^{+}	0.9748±0.0039^{+}	0.9591±0.0012^{+}	0.9669±0.0003^{+}	0.9426±0.0034^{+}	0.9943±0.0070$^{\approx}$	0.9900±0.0065^{+}	0.9857±0.0000^{+}	0.9857±0.0000^{+}	0.9857±0.0000^{+}
Adenoma	**0.9996±0.0002**	**0.9996±0.0001**$^{\approx}$	0.9958±0.0081^{+}	0.9410±0.0011^{+}	0.9783±0.0143^{+}	0.9665±0.0108^{+}	0.9461±0.0014^{+}	0.9527±0.0003^{+}	0.9287±0.0002^{+}	0.9800±0.0131^{+}	0.9743±0.0086^{+}	0.9714±0.0000^{+}	0.9714±0.0000^{+}	0.9714±0.0000^{+}
BASEHOCK	0.9113±0.0120	0.8997±0.0114^{+}	0.8597±0.0062^{+}	0.8518±0.0063^{+}	0.8715±0.0095^{+}	0.8637±0.0050^{+}	0.8860±0.0065^{+}	0.8479±0.0062^{+}	0.8597±0.0036^{+}	0.8847±0.0066^{+}	0.8861±0.0045^{+}	**0.9238±0.0082**$^{-}$	0.9150±0.0086$^{\approx}$	0.8920±0.0034^{+}
COIL20	0.9939±0.0011	0.9919±0.0008^{+}	0.9619±0.0007^{+}	0.9634±0.0017^{+}	0.9894±0.0018^{+}	0.9782±0.0006^{+}	0.9763±0.0017^{+}	0.9707±0.0010^{+}	0.9573±0.0006^{+}	0.9960±0.0009^{-}	0.9960±0.0005^{-}	**0.9971±0.0007**$^{-}$	0.9958±0.0010^{-}	0.9964±0.0009^{-}
Colon	**0.9500±0.0075**	0.9347±0.0094^{+}	0.9211±0.0092^{+}	0.8548±0.0276^{+}	0.9221±0.0203^{+}	0.8880±0.0090^{+}	0.8867±0.0110^{+}	0.8636±0.0101^{+}	0.8529±0.0059^{+}	0.9033±0.0072^{+}	0.8988±0.0071^{+}	0.8731±0.0097^{+}	0.8745±0.0145^{+}	0.8806±0.0104^{+}
DLBCL	**0.9995±0.0002**	0.9993±0.0004$^{\approx}$	0.9902±0.0016^{+}	0.9483±0.0058^{+}	0.9819±0.0092^{+}	0.9819±0.0048^{+}	0.9629±0.0055^{+}	0.9655±0.0063^{+}	0.9435±0.0069^{+}	0.9974±0.0052$^{\approx}$	0.9960±0.0061$^{\approx}$	0.9781±0.0060^{+}	0.9781±0.0060^{+}	0.9872±0.0058^{+}
Gastric	**0.9782±0.0034**	0.9677±0.0093^{+}	0.9748±0.0088$^{\approx}$	0.8955±0.0786^{+}	0.9191±0.0340^{+}	0.8874±0.0087^{+}	0.8234±0.0191^{+}	0.8564±0.0087^{+}	0.8046±0.0093^{+}	0.8908±0.0046^{+}	0.8846±0.0077^{+}	0.8154±0.0069^{+}	0.8169±0.0083^{+}	0.8215±0.0102^{+}
Gastric1	**0.9800±0.0003**	0.9738±0.0004^{+}	0.9642±0.0025^{+}	0.9310±0.0155^{+}	0.9597±0.0032^{+}	0.9496±0.0024^{+}	0.9348±0.0030^{+}	0.9402±0.0018^{+}	0.9145±0.0002^{+}	0.9588±0.0021^{+}	0.9581±0.0000^{+}	0.9588±0.0021^{+}	0.9554±0.0034^{+}	0.9588±0.0021^{+}
Gastric2	**0.9981±0.0029**	0.9936±0.0029^{+}	0.9872±0.0038^{+}	0.9522±0.0170^{+}	0.9837±0.0049^{+}	0.9710±0.0002^{+}	0.9532±0.0038^{+}	0.9623±0.0001^{+}	0.9357±0.0034^{+}	0.9837±0.0000^{+}	0.9838±0.0002^{+}	0.9765±0.0025^{+}	0.9765±0.0025^{+}	0.9832±0.0025^{+}
ISOLET	0.8894±0.0056	0.8898±0.0090$^{\approx}$	0.8557±0.0063^{+}	0.8501±0.0067^{+}	0.8822±0.0109$^{\approx}$	0.8584±0.0043^{+}	0.8903±0.0076$^{\approx}$	0.8468±0.0070^{+}	0.8607±0.0027^{+}	0.8883±0.0032$^{\approx}$	0.8874±0.0066$^{\approx}$	**0.9090±0.0059**$^{-}$	0.8963±0.0059^{-}	0.8828±0.0036^{+}
Leukaemia	**0.9995±0.0002**	0.9993±0.0002$^{\approx}$	0.9895±0.0017^{+}	0.9534±0.0015^{+}	0.9886±0.0059^{+}	0.9775±0.0059^{+}	0.9587±0.0017^{+}	0.9682±0.0037^{+}	0.9416±0.0002^{+}	0.9886±0.0057^{+}	0.9886±0.0057^{+}	0.9857±0.0000^{+}	0.9857±0.0000^{+}	0.9857±0.0000^{+}
Lymphoma	**0.9994±0.0003**	**0.9994±0.0004**$^{\approx}$	0.9868±0.0102^{+}	0.8731±0.0239^{+}	0.9404±0.0306^{+}	0.9206±0.0098^{+}	0.8841±0.0092^{+}	0.8861±0.0139^{+}	0.8585±0.0081^{+}	0.9178±0.0102^{+}	0.9200±0.0109^{+}	0.8844±0.0133^{+}	0.8867±0.0067^{+}	0.8911±0.0067^{+}
Myeloma	**0.9155±0.0101**	0.8971±0.0072^{+}	0.8825±0.0084^{+}	0.8450±0.0244^{+}	0.8702±0.0069^{+}	0.8550±0.0079^{+}	0.8522±0.0071^{+}	0.8304±0.0077^{+}	0.8186±0.0056^{+}	0.8535±0.0037^{+}	0.8529±0.0060^{+}	0.8594±0.0070^{+}	0.8482±0.0115^{+}	0.8425±0.0035^{+}
PCMAC	**0.8705±0.0139**	0.8424±0.0163^{+}	0.8007±0.0051^{+}	0.7870±0.0080^{+}	0.8103±0.0090^{+}	0.8036±0.0095^{+}	0.8256±0.0095^{+}	0.7840±0.0058^{+}	0.7982±0.0062^{+}	0.8175±0.0056^{+}	0.8105±0.0086^{+}	0.8523±0.0083^{+}	0.8505±0.0117^{+}	0.8169±0.0035^{+}
Prostate	**0.9701±0.0032**	0.9595±0.0045^{+}	0.9638±0.0049^{+}	0.8458±0.0083^{+}	0.8887±0.0092^{+}	0.8721±0.0088^{+}	0.8711±0.0091^{+}	0.8526±0.0102^{+}	0.8485±0.0038^{+}	0.8840±0.0031^{+}	0.8839±0.0054^{+}	0.8704±0.0120^{+}	0.8703±0.0107^{+}	0.8743±0.0052^{+}
SRBCT	0.9986±0.0004	0.9984±0.0006$^{\approx}$	0.9944±0.0043^{+}	0.9321±0.0085^{+}	0.9692±0.0148^{+}	0.9695±0.0091^{+}	0.9659±0.0048^{+}	0.9515±0.0116^{+}	0.9488±0.0050^{+}	**1.0000±0.0000**$^{-}$	**1.0000±0.0000**$^{-}$	0.9853±0.0074^{+}	0.9829±0.0099^{+}	0.9829±0.0080^{+}
T1D	**0.8598±0.0094**	0.8232±0.0126^{+}	0.8246±0.0084^{+}	0.7679±0.0461^{+}	0.7610±0.0169^{+}	0.7318±0.0065^{+}	0.7237±0.0049^{+}	0.7113±0.0064^{+}	0.7063±0.0041^{+}	0.7256±0.0049^{+}	0.7227±0.0053^{+}	0.7118±0.0063^{+}	0.7088±0.0088^{+}	0.7179±0.0048^{+}
madelon	**0.8822±0.0180**	0.8659±0.0246$^{\approx}$	0.7679±0.0121^{+}	0.7521±0.0095^{+}	0.7963±0.0165^{+}	0.7669±0.0229^{+}	0.8106±0.0089^{+}	0.7497±0.0169^{+}	0.7652±0.0071^{+}	0.7920±0.0049^{+}	0.7845±0.0048^{+}	0.8214±0.0097^{+}	0.8122±0.0055^{+}	0.7808±0.0071^{+}
Average	**0.9574**	0.9488	0.9325	0.8868	0.9177	0.9021	0.8944	0.8839	0.8698	0.9132	0.9110	0.9065	0.9042	0.9009

Bold indicates the best accuracy. +, ≈, and − denote that KDS-EL is significantly superior, statistically equivalent, or inferior ($p < 0.05$, paired Wilcoxon signed-rank test), respectively.

Table 5: Comparison of the number of selected features.

Dataset	KDS-EL	w/o TPS	w/o KER	MEL	FWPSO	PSO-CSM	SaWDE	IGPSO	BPSO	W-MPSO	fMPSO	NSGA-II	MOEA/D	MOPSO
ALL1	**39**	55	55	4248	612	1206	3667	2501	5886	2333	2339	5368	5548	5920
ALL4	124	152	**64**	1082	1326	1447	4581	2677	6217	3645	3213	5461	5736	6126
ALL_AML	**67**	70	473	2369	521	949	1996	1444	3266	1678	1562	2872	2968	3266
Adenoma	66	**33**	120	2483	278	937	2102	1608	3399	1691	1551	3000	3127	3420
BASEHOCK	**134**	208	812	1965	2066	1113	1912	1894	2419	1390	1617	2073	2150	2319
COIL20	84	**60**	348	323	90	167	232	255	407	405	407	283	286	405
Colon	**14**	35	105	702	240	284	621	448	935	580	612	690	730	903
DLBCL	64	**52**	615	2784	876	1074	2155	1718	3379	1963	2067	2904	3010	3349
Gastric	100	97	**62**	3604	1070	2782	7072	4916	11004	5744	6082	10165	10342	10938
Gastric1	**31**	217	678	7393	1420	2966	6963	5074	10661	6148	6044	9954	10128	10706
Gastric2	**83**	140	444	7084	1362	3062	6708	5134	10627	5803	5956	9919	10062	10701
ISOLET	**60**	100	225	250	172	194	213	195	272	242	255	185	174	256
Leukaemia	66	**47**	488	2409	534	934	2026	1440	3246	1492	1503	2893	2973	3272
Lymphoma	45	**23**	50	1404	304	524	1204	880	1832	1027	1076	1536	1609	1834
Myeloma	250	365	**175**	2137	1863	1847	4543	2859	6259	3854	3866	5585	5839	6113
PCMAC	**91**	329	637	1339	1209	651	1292	1137	1635	966	912	1352	1431	1560
Prostate	26	**23**	27	5058	1898	1541	4229	2729	6211	3294	3384	5532	5676	6100
SRBCT	64	**36**	129	906	431	366	739	515	1083	700	742	818	874	1044
T1D	330	366	**174**	6715	4984	9047	18009	13170	26971	16096	16774	25649	25939	26830
madelon	**18**	26	130	190	122	95	162	171	221	149	174	128	132	204
Average	**87.8**	121.7	290.6	2722.2	1068.9	1559.3	3521.3	2538.2	5296.5	2960.0	3006.8	4818.4	4936.7	5263.3

order-of-magnitude reduction in feature redundancy. On near-perfect saturated datasets such as *ALL1* and *SRBCT*, KDS-EL attains 0.9998 and 0.9986 accuracy with minimal signatures of 39 and 22 features. Furthermore, KDS-EL demonstrates remarkably low run-to-run dispersion, exhibiting a standard deviation of only 0.0089 on *Gastric1*, compared with 0.0215 for FWPSO and 0.0381 for MEL. This confirms that hybrid MI initialization and knowledge-guided elite refinement not only deliver higher accuracy in finding optimal biomarker subsets, but also substantially enhance the stability and robustness of the algorithm.

4.2.2. Comparison of the number of selected features

The KDS-EL framework was evaluated against 13 comparison algorithms across 20 high-dimensional benchmark datasets. Table 5 presents the experimental results for the average number of selected features and the feature selection ratio, with the optimal values in bold.

First, comparing the KDS-EL algorithm to the 13 comparison algorithms overall, KDS-EL achieves the most compact feature subsets across the entire benchmark suite, selecting an overall mean of only 87.8 features and compressing the search space down to approximately 2%–3% of its original dimension. This extreme dimensionality reduction capability is especially pronounced on challenging biomedical microarrays such as *ALL1*, *Colon*, *Gastric1*, and *Prostate*, where KDS-EL isolates minimal biomarker signatures of fewer than 40 genes, notably 14 on *Colon* and 26 on *Prostate*, while delivering near-perfect classification accuracies between 0.9500 and 0.9998. Secondly, compared to specialized single-objective evolutionary algorithms, even the most compact competitor, FWPSO, retains an average of 1,068.9 features—over 12 times the subset size of KDS-EL—demonstrating that Two-Phase Weight Scheduling successfully enforces aggressive feature elimination during early iterations. Against the original MEL baseline with 2,722.2 features, KDS-EL achieves a remarkable 31-fold reduction in feature count, confirming that mutual information initialization and knowledge-guided elite refinement effectively steer particles into highly sparse, discriminative subspaces rather than dense noisy regions. Furthermore, while multi-objective paradigms like NSGA-II-FS and fMOPSO-FS attain competitive accuracy on certain benchmarks, their operational solutions extracted via Eq. (24) retain extensive subsets ranging from 2,960.0 to 5,263.3 features. This behavior arises because unguided Pareto dominance in ultra-high dimensions exerts weak selection pressure toward extreme feature sparsity; without domain guidance, non-dominated candidate masks containing thousands of noisy dimensions easily retain parity in training-fold accuracy, clustering the minimum-error region of the Pareto front within overly dense subspaces. Overall, KDS-EL discovers substantially more compact feature subsets while simultaneously securing superior classification accuracy.

From the perspective of algorithmic components, the step-wise performance progression reveals an aggregate, macro-level compression in feature redundancy across the benchmark suite. Averaged over all 20 benchmark datasets, the feature count contracts from 2,722.2 in baseline MEL to 290.6 with mutual information seeding, further shrinks to 121.7 through knowledge-guided elite refinement, and ultimately settles at an overall compact signature of 87.8 features under the dynamic TPS strategy. This hierarchical reduction demonstrates that each proposed mechanism eliminates non-informative dimensions in a complementary manner, establishing clear overall feature-compression superiority over existing evolutionary wrappers.

Crucially, while feature reduction progresses consistently in terms of the multi-dataset average, subset cardinality on individual problem instances reflects a deliberate, domain-specific trade-off governed by the two-phase scheduling dynamics. On several challenging biomedical microarrays, such as

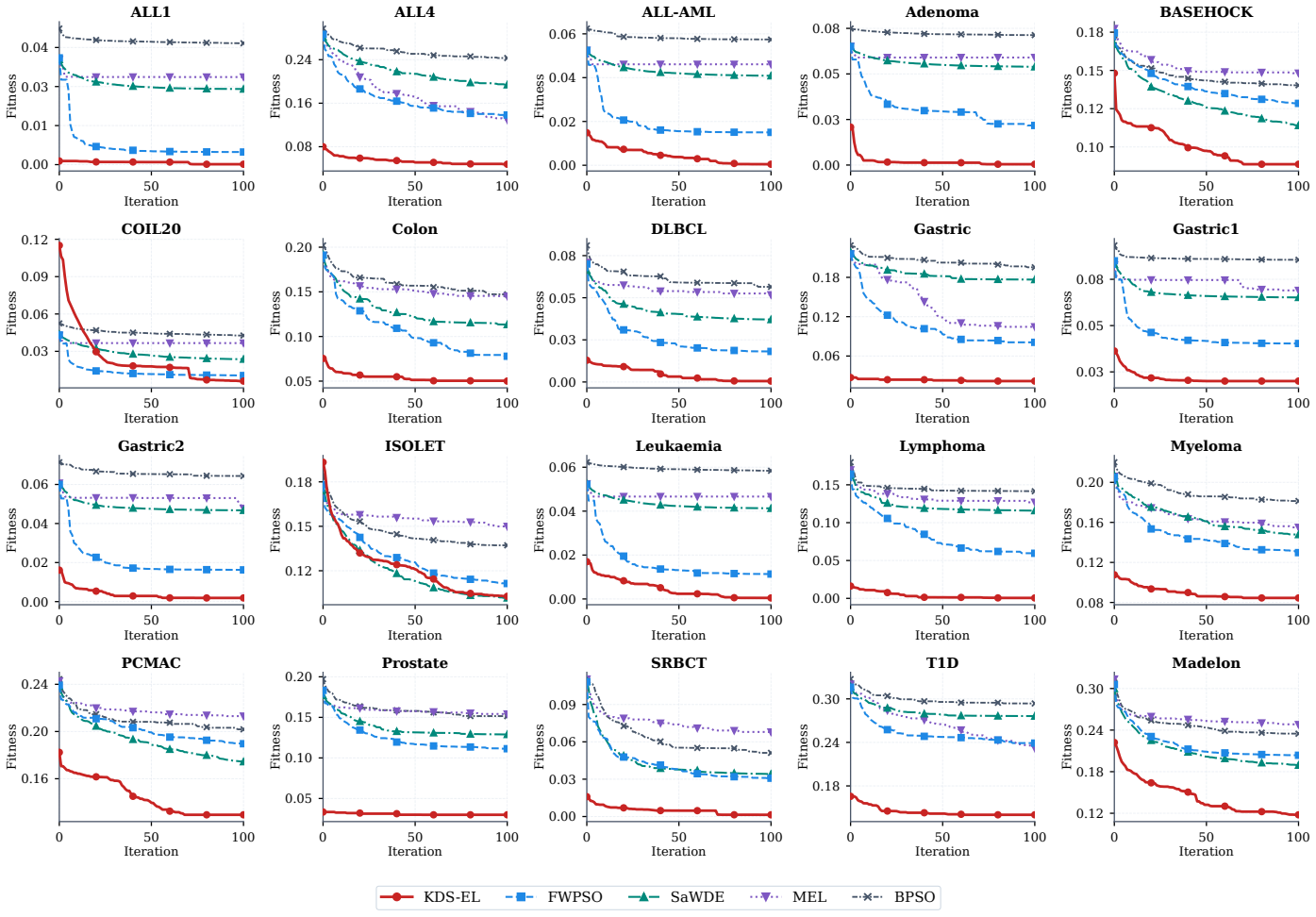


Figure 5: Average fitness convergence curves of KDS-EL and competitor algorithms.

Adenoma, *Lymphoma*, and *SRBCT*, the complete KDS-EL framework retains moderately more features than the static-weight variant without TPS. This behavior directly manifests the underlying design rationale of the two-phase strategy: upon transitioning to Phase 2, the feature cardinality penalty $w_{f,2} = 0.05$ is relaxed while the classification accuracy weight $w_{a,2} = 0.95$ is prioritized. Rather than enforcing an inflexible sparsity constraint that risks pruning decisive biomarker interactions, this controlled relaxation allows the search engine to strategically retrieve a small subset of weakly individual but strongly synergistic features that resolve ambiguous classification boundaries. Consequently, this targeted trade-off converts marginal feature retention into a statistically confirmed accuracy advantage across the benchmark suite ($p = 7.63 \times 10^{-5}$ under the Holm–Bonferroni procedure), confirming that KDS-EL balances extreme feature sparsity with optimal predictive discrimination.

4.2.3. *Fitness value and convergence analysis*

The scalar fitness value formalizes the joint optimization trade-off between minimizing classification error and compressing the active feature ratio throughout search, directly capturing each algorithm's exploratory efficiency and exploitation depth. To evaluate convergence dynamics comprehensively and avoid selective benchmark bias, Fig. 5 plots the mean fitness trajectories across all 20 benchmark datasets over 100 iterations for KDS-EL and four representative single-objective wrapper competitors: FWPSO, SaWDE, MEL, and classical BPSO. Multi-objective evolutionary algorithms (such as NSGA-II-FS, MOEA/D-FS, and W-MOPSO-FS) simultaneously evolve non-dominated Pareto fronts rather than minimizing a fixed scalarized fitness function; their multi-objective trade-offs are evaluated independently in Section 4.2.1, Section 4.2.2, and Section 4.2.6.

As demonstrated across the 20 panels in Fig. 5, KDS-EL establishes a decisive convergence advantage, achieving the lowest final fitness values across virtually all benchmark datasets. In the initial generation, mutual information soft prior initialization seeds the swarm directly in highly compact, discriminative subspaces. Consequently, KDS-EL commences search from an initial fitness baseline substantially lower than those of competing methods, which start above 0.15 to 0.30 due to unguided random bit assignments that over-select thousands of noisy features. Throughout early iterations ($t \leq 20$), KDS-EL exhibits a steep, monotonic descent across both biomedical microarrays (such as *ALL4*, *Colon*, and *Gastric*) and large-scale vision/speech datasets (such as *BASEHOCK* and *COIL20*), rapidly purging non-informative dimensions while competitors struggle to escape dense initial configurations.

The comparative trajectories further reveal distinct search stagnation patterns among the baseline algorithms. Standard BPSO exhibits erratic, sluggish progress across high-dimensional landscapes, frequently stagnating after generation 20 due to uniform velocity clipping and the absence of domain guidance. While FWPSO and SaWDE demonstrate progressive fitness improvements, their trajectories level off prematurely between generations 40 and 60 because memoryless stochastic perturbations fail to reinforce synergistic feature combinations. Similarly, baseline MEL suffers from an extended exploratory plateau during early generations before locating viable descent directions. In contrast, KDS-EL sustains continuous fitness reduction up to generation 100. By dynamically accumulating historical feature-importance feedback within the importance memory and steering elite flips toward high-utility candidates, KDS-EL continually escapes local attractors, securing an order-of-magnitude fitness reduction on ultra-high-dimensional benchmarks such as *ALL1*, *DLBCL*, *Leukaemia*, *Lymphoma*, and *T1D*.

4.2.4. *Statistical significance analysis via Wilcoxon test*

To rigorously evaluate whether the empirical performance advantages of KDS-EL are statistically meaningful rather than artifacts of stochastic variation, we follow the multi-problem non-parametric statistical testing framework recommended by Demšar [39]. Table 6 presents the pairwise Wilcoxon signed-rank test results at the $\alpha = 0.05$ significance level across the 20 benchmark datasets. As reported in Table 6, KDS-EL achieves statistically significant superiority over MEL, PSO-CSM, IGPSO, and BPSO across all 20 datasets without a single loss, and significantly surpasses KDS-EL w/o TPS&KER, FWPSO, and SaWDE on 19 datasets. Compared with KDS-EL w/o TPS, KDS-

Table 6: Pairwise Wilcoxon signed-rank test results on classification accuracy.

Method	KDS-EL	w/o TPS	w/o KER	MEL	FWPSO	PSO-CSM	SaWDE	IGPSO	BPSO	W-MPSO	fMPSO	NSGA-II	MOEA/D	MOPSO
KDS-EL	—	13/20	19/20	20/20	19/20	20/20	19/20	20/20	20/20	14/20	15/20	16/20	16/20	18/20
KDS-EL w/o TPS	**0/20**	—	15/20	20/20	20/20	20/20	19/20	20/20	20/20	14/20	15/20	15/20	15/20	17/20
KDS-EL w/o TPS&KER	**0/20**	0/20	—	18/20	11/20	15/20	15/20	19/20	16/20	10/20	9/20	14/20	14/20	13/20
MEL	**0/20**	0/20	0/20	—	0/20	1/20	3/20	3/20	11/20	2/20	2/20	3/20	3/20	3/20
FWPSO	**0/20**	0/20	0/20	0/20	—	16/20	15/20	20/20	20/20	6/20	7/20	9/20	10/20	9/20
PSO-CSM	**0/20**	0/20	0/20	0/20	0/20	—	12/20	20/20	16/20	2/20	2/20	5/20	5/20	6/20
SaWDE	**0/20**	0/20	0/20	0/20	0/20	0/20	—	10/20	20/20	2/20	2/20	3/20	3/20	6/20
IGPSO	**0/20**	0/20	0/20	0/20	0/20	0/20	0/20	—	14/20	0/20	0/20	2/20	1/20	2/20
BPSO	**0/20**	0/20	0/20	0/20	0/20	0/20	0/20	0/20	—	0/20	0/20	0/20	0/20	0/20
W-MOPSO-FS	**0/20**	0/20	0/20	0/20	0/20	0/20	0/20	0/20	0/20	—	2/20	10/20	12/20	12/20
fMOPSO-FS	**0/20**	0/20	0/20	0/20	0/20	0/20	0/20	0/20	0/20	0/20	—	10/20	10/20	8/20
NSGA-II-FS	**0/20**	0/20	0/20	0/20	0/20	0/20	0/20	0/20	0/20	0/20	0/20	—	6/20	5/20
MOEA-D-FS	**0/20**	0/20	0/20	0/20	0/20	0/20	0/20	0/20	0/20	0/20	0/20	0/20	—	4/20
MOPSO-FS	**0/20**	0/20	0/20	0/20	0/20	0/20	0/20	0/20	0/20	0/20	0/20	0/20	0/20	—

Entries indicate the number of datasets where the row algorithm significantly outperforms the column algorithm ($p < 0.05$).

EL records 13 wins and 7 ties, confirming the significant contribution of Two-Phase Weight Scheduling. Furthermore, against the five multi-objective methods, KDS-EL secures 14 to 18 wins. Statistical ties occur exclusively on ceiling-effect datasets where classification accuracy saturates, while KDS-EL retains substantial advantages in feature compactness. Crucially, no competing algorithm achieves statistical superiority over KDS-EL on any dataset.

To control the family-wise error rate across multiple pairwise comparisons, the Holm–Bonferroni step-down procedure [41] is applied to the $k = 13$ hypothesis tests with KDS-EL serving as the control method, with the detailed statistical outcomes summarized in Table 7. Following the multi-dataset non-parametric testing framework of Demšar [39], the pairwise Wilcoxon signed-rank test is performed on the paired mean classification accuracy values across the $N = 20$ benchmark datasets. Table 7 details the mean accuracy advantage ΔAcc, the rank-sum statistic $W = \min(R^+, R^-)$, the standardized normal statistic Z, the exact two-sided p-value, and the sequential Holm critical threshold $\alpha/(k - i + 1)$ at the family-wise significance level $\alpha = 0.05$. Across all 13 pairwise comparisons, the computed exact p-values range from 1.91×10^{-6} to 2.10×10^{-4}, each remaining strictly below its respective Holm-adjusted threshold ($\alpha_{\text{Holm}} \in [0.0038, 0.0500]$). Consequently, all 13 null hypotheses of equivalent performance are formally rejected, confirming that the empirical superiority of KDS-EL over both single-objective and multi-objective competitors is statistically rigorous and reproducible under multi-dataset benchmarking standards.

Table 7: Holm–Bonferroni procedure for pairwise statistical comparisons.

Method	ΔAcc	W	Z	p	α_{Holm}	Reject H_0
BPSO	+0.0875	0.0	−3.92	1.91×10^{-6}	0.0038	✓
IGPSO	+0.0734	0.0	−3.92	1.91×10^{-6}	0.0042	✓
MEL	+0.0706	0.0	−3.92	1.91×10^{-6}	0.0045	✓
PSO-CSM	+0.0552	0.0	−3.92	1.91×10^{-6}	0.0050	✓
FWPSO	+0.0396	0.0	−3.92	1.91×10^{-6}	0.0056	✓
KDS-EL w/o TPS&KER	+0.0248	0.0	−3.92	1.91×10^{-6}	0.0063	✓
SaWDE	+0.0630	1.0	−3.88	3.81×10^{-6}	0.0071	✓
MOPSO-FS	+0.0565	3.0	−3.81	9.54×10^{-6}	0.0083	✓
fMOPSO-FS	+0.0464	7.0	−3.66	3.62×10^{-5}	0.0100	✓
W-MOPSO-FS	+0.0442	8.0	−3.62	4.77×10^{-5}	0.0125	✓
MOEA-D-FS	+0.0532	9.0	−3.58	6.29×10^{-5}	0.0167	✓
KDS-EL w/o TPS	+0.0086	5.0	−3.51	7.63×10^{-5}	0.0250	✓
NSGA-II-FS	+0.0509	14.0	−3.40	2.10×10^{-4}	0.0500	✓

Note: ΔAcc denotes the mean accuracy advantage of KDS-EL over each competitor. $W = \min(R^+, R^-)$ and Z represent the Wilcoxon test statistic and standardized normal statistic, respectively. p is the exact two-sided p-value, and $\alpha_{\text{Holm}} = \alpha/(k - i + 1)$ is the Holm-adjusted significance threshold ($\alpha = 0.05, k = 13$).

To establish comprehensive global algorithmic rankings, the non-parametric Friedman test [40] is conducted across all 20 benchmark datasets, as depicted in Fig. 6. KDS-EL secures the lowest average Friedman rank of 1.90, followed by KDS-EL w/o TPS with a rank of 2.95. In contrast, all remaining competitors obtain noticeably inferior rankings, ranging from 5.33 for W-MOPSO-FS to 13.80 for standard BPSO. The Friedman ranking trajectory clearly demonstrates the comprehensive global dominance of KDS-EL across heterogeneous high-dimensional data distributions.

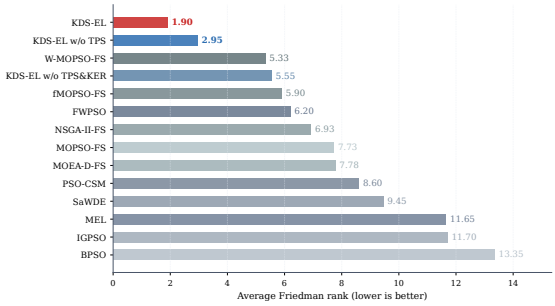


Figure 6: Average Friedman ranks of the compared algorithms.

4.2.5. *Performance comparison before and after algorithm improvement*

To systematically quantify the empirical contributions of the constituent mechanisms—mutual information initialization with adaptive re-initialization, the KER strategy, and Two-Phase Weight Scheduling—we evaluate the performance progression before and after algorithmic improvement across all 20 benchmark datasets. Following the subtractive analysis paradigm in evolutionary computation [18, 19], the complete KDS-EL framework is evaluated against its sequentially stripped configurations. Table 8 summarizes the aggregate performance progression across the 20 benchmark datasets, Fig. 7 visualizes the multi-dataset classification accuracy contours in a radar chart, and Fig. 8 tracks the evolutionary accuracy convergence dynamics across the benchmark suite. Complete KDS-EL, which achieves a mean accuracy of 0.9574 with 87.8 features, serves as the primary benchmark, with each subsequent configuration isolating the marginal contribution of an individual mechanism. Granular per-dataset ablation metrics across all 20 benchmarks are documented in Supplementary Table S4.

Table 8: Results of performance comparison before and after algorithm improvement.

Variant	Accuracy	Features	ΔAcc (pp)
KDS-EL	**0.9574**	**87.8**	–
KDS-EL w/o TPS	0.9488	122	-0.86
KDS-EL w/o TPS&KER	0.9325	291	-1.63
MEL	0.8868	2722	-4.58

ΔAcc indicates the marginal accuracy degradation when stripping the corresponding mechanism.

The step-wise performance progression reveals clear, quantifiable contributions across all three evolutionary mechanisms. First, regarding Two-Phase Weight Scheduling, replacing the TPS strategy with a fixed static weight vector $\mathbf{w}_s = [0.90, 0.10]$ results in an immediate accuracy drop of 0.86% and expands the selected subset from 87.8 to 121.7 features, while increasing runtime from 51.4 to 53.6 seconds. The dynamical necessity of TPS is vividly corroborated by the convergence trajectories in Fig. 8. During Phase 1 ($t \leq 70$), the aggressive feature-size penalty $\mathbf{w}_1 = [0.70, 0.30]$ enforces rigorous dimensional pruning, during which KDS-EL maintains accuracy parity with KDS-EL

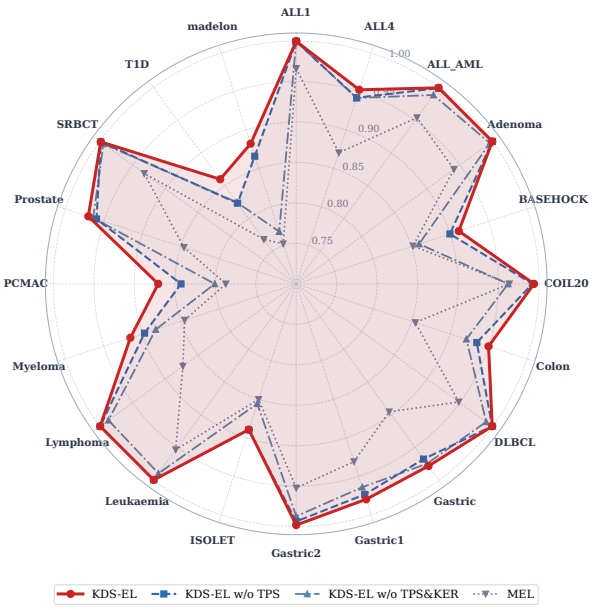


Figure 7: Radar charts of classification accuracy for different algorithm variants.

w/o TPS while selecting substantially fewer features. Exactly upon crossing the scheduled boundary at $t = 70$, switching to accuracy-prioritized weights $\mathbf{w}_2 = [0.95, 0.05]$ produces a distinct secondary accuracy acceleration visible in the overall mean trajectory of Fig. 8(a) and in challenging benchmarks such as *BASEHOCK*, *COIL20*, *T1D*, and *Madelon*. In contrast, the static variant KDS-EL w/o TPS plateaus prematurely after generation 30, failing to achieve late-stage boundary refinement.

Second, regarding Knowledge-Guided Elite Refinement, removing the KER strategy and relying solely on standard particle swarm updates inflicts an additional accuracy loss of 1.63%, falling to 0.9325, while more than doubling the retained feature count to 290.5. As evidenced across all panels of Fig. 8, stripping KER depresses the convergence envelope throughout the entire evolutionary timeline, demonstrating that closed-loop feature-importance accumulation is essential for steering elite mutations toward productive dimensions. Third, regarding Mutual Information Initialization and Adaptive Re-initialization, stripping the MI seeding and stagnation-triggered re-initialization mechanisms—reducing the framework to the unguided multi-task swarm MEL [12] initialized uniformly at random—triggers a severe performance drop: classification accuracy falls by 4.58% to 0.8868, and the mean feature count surges to 2,722.2. The convergence curves in Fig. 8 confirm that unguided random initialization wastes dozens of generations wandering in dense, non-informative subspaces before making meaningful progress.

Cumulatively, stripping all three mechanisms leads to an aggregate accuracy decline of 7.06% and a 31-fold surge in feature subset size, from 87.8 to 2,722.2 features. This confirms that the three mechanisms operate orthogonally and in a complementary manner: MI initialization provides foundational dimensionality reduction, KER delivers fine-grained elite refinement, and TPS harmonizes search objectives across evolutionary stages. Furthermore, the structural design of greedy 1-to-1 in-place elite replacement prevents excessive selection pressure that would otherwise collapse swarm diversity, ensuring robust convergence across heterogeneous problem domains.

4.2.6. Computational efficiency and runtime analysis

Computational efficiency is a critical evaluation criterion for high-dimensional feature selection, particularly in wrapper-based metaheuristics where repeated classifier evaluations incur substantial runtime overhead. Fig. 9 visualizes the multi-objective Pareto trade-off between classification accuracy and the number of selected features across the 20 benchmark datasets, alongside runtime indicators. The complete per-dataset runtime measurements for all 14 algorithms are documented in Supplementary Table S5.

As shown in Fig. 9 and Supplementary Table S5, KDS-EL achieves an average execution time of 51.4 seconds across the 20 benchmark datasets, outperforming 12 of the 13 competing methods. Compared with the original MEL algorithm, which requires 175.0 seconds on average, KDS-EL delivers a 3.4-fold acceleration. Furthermore, all five multi-objective methods require between 110.4 and 142.8 seconds on average, which is more than double the runtime of KDS-EL. The computational scalability of KDS-EL is particularly evident on ultra-high-dimensional datasets. On the 54,675-dimensional *T1D* dataset, KDS-EL completes in 118.8 seconds, running more than twice as fast as PSO-CSM at 256.6 seconds and SaWDE at 216.1 seconds. Similarly, on *BASEHOCK*, KDS-EL finishes in 130.1 seconds, achieving an acceleration of more than 4.7-fold over SaWDE at 616.0 seconds and PSO-CSM at 644.7 seconds. Although single-objective FWPSO exhibits a marginally faster runtime of 49.7 seconds due to early-terminating simulated annealing heuristics, it incurs a substantial 3.97% accuracy deficit and selects over 12 times more features than KDS-EL.

The superior computational scalability of KDS-EL is primarily driven by three synergistic factors. First, in k-NN wrapper evaluations, distance computation complexity scales with the number of selected features. By aggressively reducing the active feature count from thousands to dozens during initialization and Phase 1, the per-evaluation cost of the classifier decreases by orders of magnitude. Second, because KER operates exclusively on a compact elite pool with a sparse perturbation fraction, its incremental complexity scales linearly with dimension and remains negligible compared to full population evaluations. Third, unlike multi-objective algorithms that execute computationally intensive non-dominated sorting and crowding-distance updates at every generation, KDS-EL operates with scalarized dual-phase scheduling, completely eliminating Pareto archive maintenance bottlenecks. Consequently, KDS-EL occupies the optimal Pareto corner combining top-tier classification accuracy, extreme feature sparsity, and high computational efficiency.

4.2.7. Parameter sensitivity and robustness analysis

To evaluate the algorithmic robustness of KDS-EL and justify its default parameter configurations, a one-factor-at-a-time parameter sensitivity analysis is conducted following the standard methodology in evolutionary computation [36]. The evaluation is performed across four representative benchmark datasets spanning distinct dimensionalities and sample scales: *Adenoma*, *Colon*, *ALL4*, and *BASEHOCK*. Five key hyperparameters are systematically investigated: the knowledge weight decay factor γ, the phase split ratio κ, the elite perturbation fraction ϕ, the mutual information initialization ratio ρ, and the elite pool size k_e, while all other parameters are held at their default values in Table 3. Table 9 summarizes the tested parameter ranges, recommended optimal configurations, stable operating bands, and maximum accuracy deviations across the four representative datasets. Furthermore, Fig. 10 illustrates the corresponding accuracy and feature-count trajectories alongside stable operating bands across all five parameters, with complete per-dataset matrices reported in Supplementary Table S6.

Table 9: Summary of parameter sensitivity analysis.

Parameter	Tested range	Optimal	Stable range	Max ΔAcc (%)
Knowledge decay γ	0.90–1.00	0.99	0.95–0.995	±1.2
Phase split κ	0.30–0.90	0.70	0.60–0.80	±0.7
Perturbation fraction ϕ	0.005–0.10	0.01	0.005–0.02	±1.3
MI init ratio ρ	0.0–1.0	0.50	0.25–0.75	±1.2
Elite pool size k_e	1–10	3	2–5	±0.9

Max ΔAcc denotes the maximum accuracy variation across the investigated parameter range.

The sensitivity trajectories across the five hyperparameters yield critical insights into the search dynamics and stability of KDS-EL. First, regarding the knowledge weight decay factor γ, setting $\gamma = 0.99$ yields optimal performance with an average accuracy of 0.944 and 107 selected features. An overly low decay factor such as $\gamma = 0.90$ causes premature forgetting of valuable feature associations discovered in earlier iterations, degrading accuracy by 1.2%, whereas disabling decay entirely at $\gamma = 1.00$ accumulates obsolete historical feedback and inflates the feature count to 151 without accuracy gain. Second, regarding the phase split ratio κ, setting $\kappa < 0.50$ allocates insufficient iterations to Phase 1, resulting in inadequate feature pruning with 146 to 189 retained features, while an overly prolonged Phase 1 with $\kappa = 0.90$ compresses the Phase 2 refinement window, leading to a 0.7% accuracy decline. The broad range between 0.60 and 0.80 provides a stable operating regime with accuracy variations remaining strictly below 0.5%, confirming $\kappa = 0.70$ as an ideal balance. Third, regarding the elite perturbation fraction ϕ, setting $\phi = 0.01$ maintains an optimal balance between localized exploitation and exploratory mutation. Increasing $\phi \geq 0.05$ introduces excessive random perturbation that disrupts delicate elite biomarker combinations, reducing accuracy by 1.3% and doubling the selected feature count to 203. Fourth, regarding the mutual information initialization ratio ρ, disabling MI seeding entirely at $\rho = 0.0$ imposes a 1.2% accuracy penalty and increases execution time by 21% to 62.3 seconds, while setting $\rho = 1.0$ homogenizes the initial population and induces premature

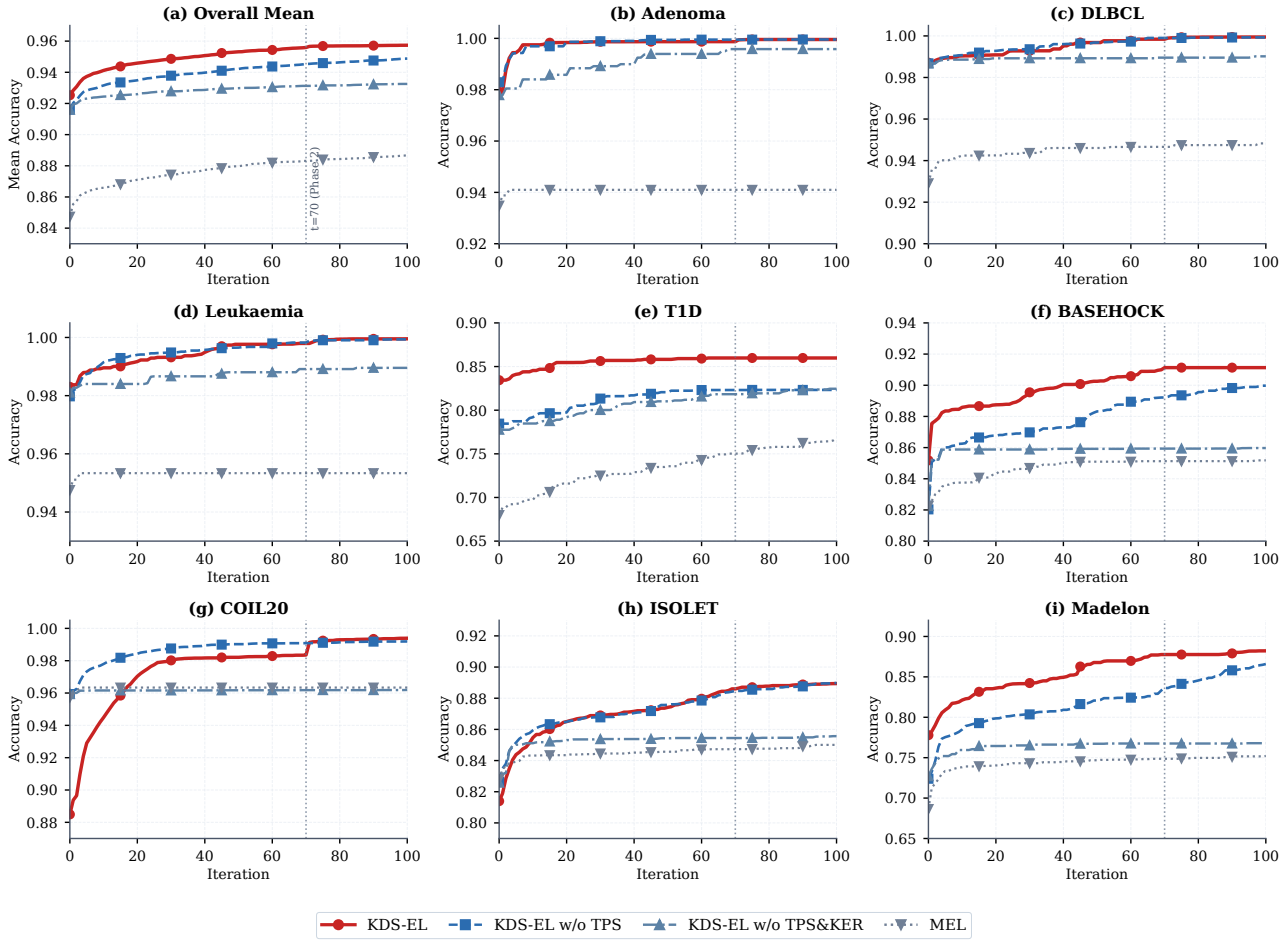


Figure 8: Convergence curves of classification accuracy for KDS-EL and its ablative variants: (a) overall mean across 20 benchmark datasets, and (b)–(i) eight representative datasets. The vertical dotted line at $t = 70$ denotes the transition boundary from Phase 1 (dimensional pruning) to Phase 2 (fine accuracy refinement) in Two-Phase Weight Scheduling.

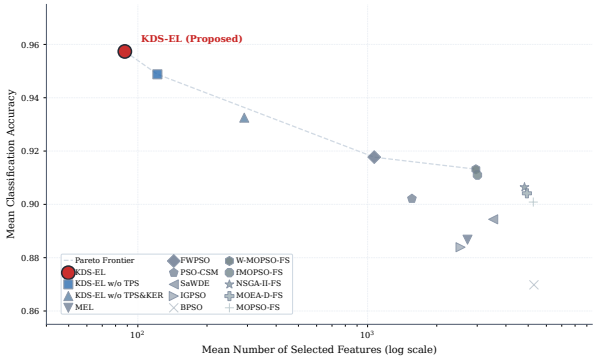


Figure 9: Multi-objective Pareto efficiency and runtime trade-off.

convergence. The default balanced setting $\rho = 0.50$ effectively reconciles prior domain knowledge guidance with stochastic diversity. Finally, regarding the elite pool size k_e, refining only a single elite with $k_e = 1$ restricts knowledge propagation across subpopulations, while setting $k_e = 10$ dilutes the quality signal of matrix $\mathbf{W}$, establishing $k_e = 3$ as the optimal default.

Across all five hyperparameters, the performance trajectories in Fig. 10 demonstrate wide, forgiving basins of attraction rather than sharp, brittle peaks. The maximum accuracy variation across the identified stable operating bands remains strictly within 1.5%, and KDS-EL consistently outperforms the comparison algorithms across broad parameter ranges. This demonstrates that KDS-EL exhibits exceptional hyperparameter robustness and plug-and-play adaptability, delivering reliable optimization performance across diverse high-dimensional applications without requiring tedious, problem-specific hyperparameter tuning.

4.2.8. *Discussion*

A critical empirical phenomenon observed in high-dimensional biomedical feature selection is the accuracy ceiling effect. Among the 20 benchmark datasets, eight gene-expression microarrays, including *ALL1*, *ALL_AML*, *Adenoma*, *DLBCL*, *Gastric2*, *Leukaemia*, *Lymphoma*, and *SRBCT*, exhibit classification accuracy exceeding 0.998 across top-performing methods. Under such near-perfect classification conditions, the few available test instances are completely separated, and accuracy differences among leading algorithms become narrower than 0.002, corresponding to stochastic variations in at most one cross-validation sample. Consequently, classification accuracy alone ceases to provide discriminative utility, and the number of selected features becomes the decisive criterion for evaluating algorithmic efficacy.

Under this saturated regime, KDS-EL demonstrates substantial superiority in feature subset compactness, identifying ultra-sparse biomarker signatures of only 39 to 83 features across all eight ceiling benchmarks. Specifically, KDS-EL extracts minimal subsets of 39 features on *ALL1*, 45 on *Lymphoma*, and 64 on *DLBCL*. In stark contrast, single-objective FWPSO requires 278 to 1,362 features to achieve comparable accuracy, multi-objective methods retain extensive subsets ranging from 700 to 10,701 features, where MOPSO-FS selects 5,920 features on *ALL1* and 10,701 features on *Gastric2*, and the original MEL algorithm retains 2,483 to 4,248 features. This demonstrates that conventional evolutionary algorithms suffer from severe feature bloat in ceiling scenarios, whereas KDS-EL effectively isolates the true minimal discriminative subset.

This dimensional contrast highlights the profound practical and clinical relevance of KDS-EL. In translational bioinformatics and cancer diagnostics, an ultra-compact signature of 40 to 80 biomarkers can be readily validated through targeted RT-qPCR assays, integrated into biological pathway enrichment analyses, and engineered into cost-effective diagnostic gene panels. In contrast, candidate signatures containing hundreds or thousands of uncurated genes are cost-prohibitive to validate clinically and introduce high risks of false discoveries driven by spurious co-expression artifacts. By reconciling top-tier classification accuracy, extreme biomarker sparsity, and computational scala-

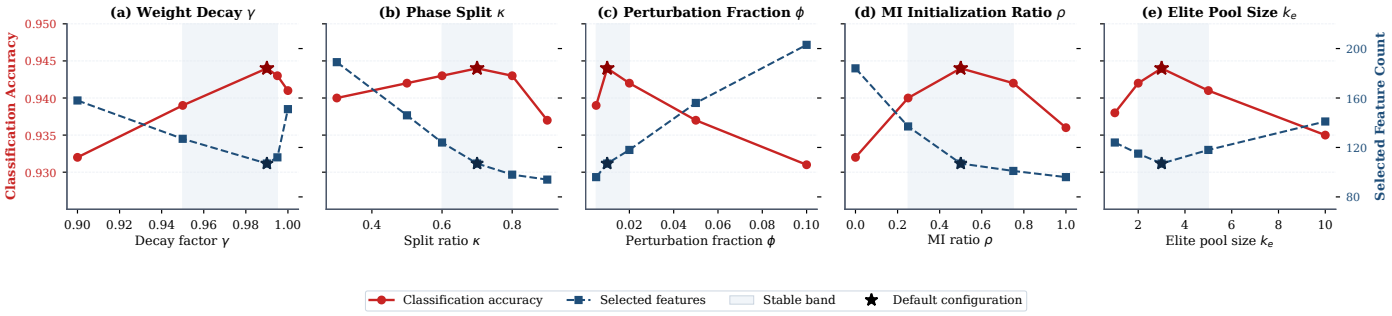


Figure 10: Parameter sensitivity analysis of KDS-EL across five key hyperparameters.

bility, KDS-EL establishes a rigorous and deployable performance frontier for high-dimensional biomedical data analytics.

Beyond nearest-neighbor metrics, a critical concern in wrapper-based evolutionary feature selection is inductive bias coupling: whether the optimized subsets merely fit the local geometric neighborhood of 3-NN while suffering performance collapse when transferred to alternative learning paradigms. To examine classifier-agnostic generalizability, the compact feature subsets discovered by KDS-EL were evaluated under Support Vector Machines with radial basis function kernels and Random Forest classifiers across representative benchmark datasets spanning distinct dimensional scales. As documented in Supplementary Table S9, the compact subsets deliver substantial classification improvements over the full-dimensional baselines across all evaluated learning architectures. On the 12,625-dimensional *ALL4* dataset, reducing the active feature space by 99.0% from 12,625 to 124 features elevates Random Forest classification accuracy from 78.36% to 90.35% and 3-NN accuracy from 69.94% to 87.25%, while SVM accuracy reaches 85.96%. Similarly, on the *Adenoma* benchmark, all three classifiers attain 100.00% classification accuracy on the 66 selected features. This cross-paradigm consistency confirms that KDS-EL extracts intrinsically salient features rather than overfitting to classifier-specific geometric manifolds.

Furthermore, translating computational feature selection into clinical insight requires that the isolated dimensions reflect genuine pathophysiological mechanisms rather than statistical noise. Granular inspection of the minimal signatures identified by KDS-EL reveals strong concordance with established cancer biology. In the *Colon* adenocarcinoma microarray, top-ranked selected features map directly to well-characterized oncological biomarkers, as detailed in Supplementary Table S10. Specifically, KDS-EL identifies *COL11A1* (probe *Hsa.36689*), a recognized driver of desmoplastic stroma and tumor invasion; *GUCA2A* (Guanylin, probe *Hsa.831*), a mucosal differentiation hormone whose loss constitutes a universal clinical hallmark of early colorectal adenoma progression [42]; and smooth muscle structural markers *DES* (Desmin, probe *Hsa.3331*) and *ACTG2* (probe *Hsa.1832*), which capture muscularis mucosae integrity versus neoplastic infiltration. In *Prostate* cancer, KDS-EL consistently isolates *HPN* (Hepsin, probe *1005_at*) and *AMACR* (Alpha-Methylacyl-CoA Racemase, probe *37639_at*), both clinical immunohistochemical gold standards in prostatic carcinoma diagnosis [43, 44]. This rigorous biological alignment corroborates that KDS-EL captures genuine molecular drivers of malignancy, bridging evolutionary machine learning with clinical translational oncology.

5. Conclusion

In this paper, the KDS-EL framework is proposed to solve the high-dimensional feature selection problem. In KDS-EL, mutual-information-guided initialization and adaptive re-initialization are first constructed to establish informative search basins and restore swarm diversity during search. In addition, to overcome unguided stochastic mutation trials, we design a knowledge-guided elite refinement strategy to accumulate cross-iteration feature-importance feedback, enhance candidate biomarker subset quality, and sustain continuous optimization momentum. Furthermore, a two-phase weight scheduling strategy temporally decouples coarse feature pruning from fine classification refinement. Based on these collaborative mechanisms, the experimental results of KDS-EL across 20 real-world high-dimensional benchmark datasets demonstrate that, compared with 13 state-of-the-art algorithms, KDS-EL removes redundant and irrelevant features more effectively, achieves superior classification accuracy, and substantially improves computational efficiency.

The knowledge-guided elite refinement strategy and stage-aware weight scheduling mechanisms in KDS-EL proposed in this paper can be used not only to improve multi-task particle swarm optimization but also to enhance other swarm intelligence and evolutionary algorithms. The newly proposed strategies enhance the optimization capabilities of metaheuristics, aiding in providing better search methodologies for the fields of machine learning and data science. For example, the improved evolutionary frameworks can be employed for optimizing neural network hyperparameters, neural architecture search, multi-omics biomarker discovery, and high-dimensional biomedical signal processing, among other tasks, thereby offering novel solutions for these domains.

Future work could improve the global optimization process of evolutionary learning by combining relevant algorithmic ideas from machine learning and deep representation learning, such as transfer learning and self-supervised feature interaction modeling, to accelerate convergence and enhance scalability on ultra-large-scale datasets. In addition, the scope of application of the algorithm can also be expanded through domain-specific research. For example, the scope of application can be explored for complex clinical challenges such as single-cell RNA sequencing, multi-label disease diagnosis, and cancer survival prediction, where domain-specific prior knowledge integration strategies can be designed to improve biomarker discovery performance and clinical interpretability in the target domain.

CRediT Author Statement

Anonymous Author: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review & editing, Visualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The 20 high-dimensional benchmark datasets analyzed in this study are publicly accessible from standard machine learning and bioinformatics repositories. The source code of KDS-EL will be made available upon publication. Supplementary experimental records—including granular per-dataset ablation metrics (Table S4), complete computational runtime measurements (Table S5), full parameter sensitivity grids (Table S6), cross-classifier inductive generalization assessments (Table S9), and biological biomarker target annotations (Table S10)—are provided in the accompanying Supplementary Material document.

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